

EG800Q Series

Hardware Design

LTE Standard Products

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Safety Information

The following safety precautions must be observed during all phases of operation, such as usage, service or repair of any terminal or mobile incorporating the module. Manufacturers of the terminal should notify users and operating personnel of the following safety information by incorporating these guidelines into all manuals of the product. Otherwise, Quectel assumes no liability for customers' failure to comply with these precautions.



Full attention must be paid to driving at all times in order to reduce the risk of an accident. Using a mobile while driving (even with a handsfree kit) causes distraction and can lead to an accident. Please comply with laws and regulations restricting the use of wireless devices while driving.



Switch off the terminal or mobile before boarding an aircraft. The operation of wireless appliances in an aircraft is forbidden to prevent interference with communication systems. If there is an Airplane Mode, it should be enabled prior to boarding an aircraft. Please consult the airline staff for more restrictions on the use of wireless devices on an aircraft.



Wireless devices may cause interference on sensitive medical equipment, so please be aware of the restrictions on the use of wireless devices when in hospitals, clinics or other healthcare facilities.



Terminals or mobiles operating over radio signal and cellular network cannot be guaranteed to connect in certain conditions, such as when the mobile bill is unpaid or the (U)SIM card is invalid. When emergency help is needed in such conditions, use emergency call if the device supports it. In order to make or receive a call, the terminal or mobile must be switched on in a service area with adequate cellular signal strength. In an emergency, the device with emergency call function cannot be used as the only contact method considering network connection cannot be guaranteed under all circumstances.



The terminal or mobile contains a transceiver. When it is ON, it receives and transmits radio frequency signals. RF interference can occur if it is used close to TV sets, radios, computers or other electric equipment.



In locations with explosive or potentially explosive atmospheres, obey all posted signs and turn off wireless devices such as mobile phone or other terminals. Areas with explosive or potentially explosive atmospheres include fueling areas, below decks on boats, fuel or chemical transfer or storage facilities, and areas where the air contains chemicals or particles such as grain, dust or metal powders.

About the Document

Revision History

Version	Date	Author	Description
-	2022-08-05	Chan Chen/ Gary Wang	Creation of the document
1.0	2023-02-07	Maffie Zhang/ Emma Wu	First official release
1.1	2023-08-02	Maffie Zhang/ Emma Wu/ Lem Jin/ Sean Fang	- Added an applicable module EG800Q-NA. - Added the information about Wi-Fi scan function (Table 3 & Figure 1). - Updated the USB serial driver; Added the baud rates and functions of auxiliary UART; Updated the maximum category supported by LTE (Table 4). - Updated the DC characteristics of STATUS, NET_STATUS, USIM_DET, MAIN_DCD, MAIN_RI and MAIN_DTR; Added the DC characteristics of PSM_IND and PSM_INT (Table 6). - Added the characteristics of WAKEUP, AGPIO and AGPIOWU pins (Table 7). - Added the baud rates and functions of auxiliary UART; Added the notifications of the connection between the module' UART and the external device (Chapter 4.3). - Updated the note of the ADC design (Chapter 4.6). - Updated the power consumption (Chapter 6.3). - Updated the digital I/O characteristics (Chapter 6.4). - Added the mounting direction (Chapter 8.3.3).
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1.4	2026-05-18	Lingeswarran A/L Sundram/ Paddy Rong/ Ooi Kuan Zhe/ Andy Peng	1. Added the applicable module EG800Q-NAD, EG800Q-GL and related information. 2. Added a note on avoiding current sink on the module's pins (Chapter 2.4). 3. Added a note that transmit power supported by the module in hardware may be subject to local regulations in actual applications (Chapter 5.1.2). 4. Updated module top views (Figure 38 & 39).
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1 Introduction

This document defines EG800Q series module and describes its air interface and hardware interfaces, which are connected with your applications.

This document provides a quick insight into EG800Q series module interface specifications, electrical and mechanical details, as well as other related information of the module. Coupled with application notes and user guides, the document makes it easy to design and set up mobile applications with the module.

1.1. Special Mark

Table 1: Special Mark

Mark	Definition
*	Unless otherwise specified, an asterisk (*) after a function, feature, interface, pin name, command, argument, and so on indicates that it is under development and currently not supported; and the asterisk (*) after a model indicates that the model sample is currently unavailable.

2 Product Overview

The module is an SMD type module.

Table 2: Basic Information

EG800Q Series	
Packaging	LGA
Pin counts	109
Dimensions	(17.7 ±0.2) mm × (15.8 ±0.2 mm) × (2.4 ±0.2) mm
Weight	approx. 2.0 g
Wireless network function	LTE/Wi-Fi Scan

2.1. Frequency Bands and Functions

Table 3: Frequency Bands and Functions Supported by EG800Q Series

Mode	EG800Q-EU	EG800Q-NA	EG800Q-IN	EG800Q-NAD	EG800Q-GL
LTE-FDD	B1/B3/B5/B7/ B8/B20/B28	B2/B4/B5/B12/ B13/B66	B1/B3/B5/B8	B2/B4/B5/B12/ B13/B25/B66/ B71	B1/B2/B3/B4/B5/B7 /B8/B12/B13/B14/ B17/B18/B19/B20/ B25/B26/B28/B66/ B71
LTE-TDD	-	-	B40/B41	-	B34/B38/B39/B40/ B41
Wi-Fi Scan	802.11b/g/n with 2.4G DSSS beacon	802.11b/g/n with 2.4G DSSS beacon	802.11b/g/n with 2.4G DSSS beacon	802.11b/g/n with 2.4G DSSS beacon	802.11b/g/n with 2.4G DSSS beacon

NOTE

1. The Wi-Fi scan function utilizes the same antenna interface as the main antenna. Due to this shared interface, Time Division Multiplexing (TDM) is employed since the two functions cannot be used simultaneously. Wi-Fi Scan functionality only supports receiving. Transmitting is not supported.
2. The module supports the above frequency bands in hardware, but actual applications may be subject to local regulations.

2.2. Key Features

Table 4: Key Features

Features	Details
Power Supply	3.3–4.3 V - Typical supply voltage: 3.8 V
SMS	- Text and PDU mode - Point-to-point MO and MT - SMS cell broadcast - SMS storage: ME by default
USIM Interface	USIM card: 1.8 V, 3.0 V
PCM Interface*	- Digital audio interface: PCM interface - Used for audio function with external Codec
I2C Interface*	- One I2C interface - Compliant with I2C bus specification
USB Interface	- Compliant with USB 2.0 specifications (only supports slave mode) - Data rate: up to 480 Mbps - Used for AT command communication, data transmission, software debugging, firmware upgrading and outputting logs - USB Serial Driver: Windows 10/11, Linux 2.6–6.7, Android 4.x–14.x
UART Interfaces	Main UART: - Used for AT command communication and data transmission - Baud rate: 115200 bps by default - RTS and CTS hardware flow control Debug UART: - Used for outputting partial logs - Baud rate: 115200 bps and 3000000 bps Auxiliary UART*: - Baud rate: 115200 bps by default

Network Indication	NET_STATUS to indicate network connectivity status
AT Commands	● Compliant with 3GPP TS 27.007, 3GPP TS 27.005 ● Compliant with Quectel enhanced AT commands
Antenna Interface	● LTE/Wi-Fi Scan antenna interface (ANT_MAIN) ● 50 Ω impedance
Transmit Power	LTE: Class 3 (23 dBm $\pm$ 2 dB)
LTE Features	● 3GPP Rel-14 FDD and TDD ● Max. LTE category: Cat 1bis ● 1.4/3/5/10/15/20 MHz RF bandwidth ● UL modulation: QPSK, 16QAM ● DL modulation: QPSK, 16QAM and 64QAM ● LTE-FDD: Max. 10 Mbps (DL)/ 5 Mbps (UL) ● LTE-TDD: Max. 8.96 Mbps (DL)/ 3.1 Mbps (UL)
Internet Protocol Features	● TCP/UDP/NTP/NITZ/FTP/HTTP/PING/HTTPS/FTPS/SSL/MQTT/CMUX/PPP/FILE/SMTP/SMTPS/MMS* protocols ● PAP and CHAP for PPP connections
Temperature Range	● Normal operating temperature ¹: -35 °C to +75 °C ● Extended operating temperature ²: -40 °C to +85 °C ● Storage temperature: -40 °C to +90 °C
Firmware Upgrade	Via USB interface or DFOTA
RoHS	All hardware components are fully compliant with EU RoHS directive

¹ Within this range, the module's indicators comply with 3GPP specification requirements.

² Within the range of -40 to -35 °C or 75 to 85 °C, the module retains the ability to establish and maintain functions such as SMS and data transmission, without any unrecoverable malfunction. Radio spectrum and radio network remain uninfluenced, whereas the value of one or more parameters, such as P_{out}, may decrease and fall below the range of the 3GPP specified tolerances. When the temperature returns to the normal operating temperature range, the module's indicators will comply with 3GPP specification requirements again.

2.3. Block Diagram

The following figure shows a block diagram of the module and illustrates the major functional parts of the module.

- Power management
- Baseband part
- Radio frequency part
- Peripheral interfaces

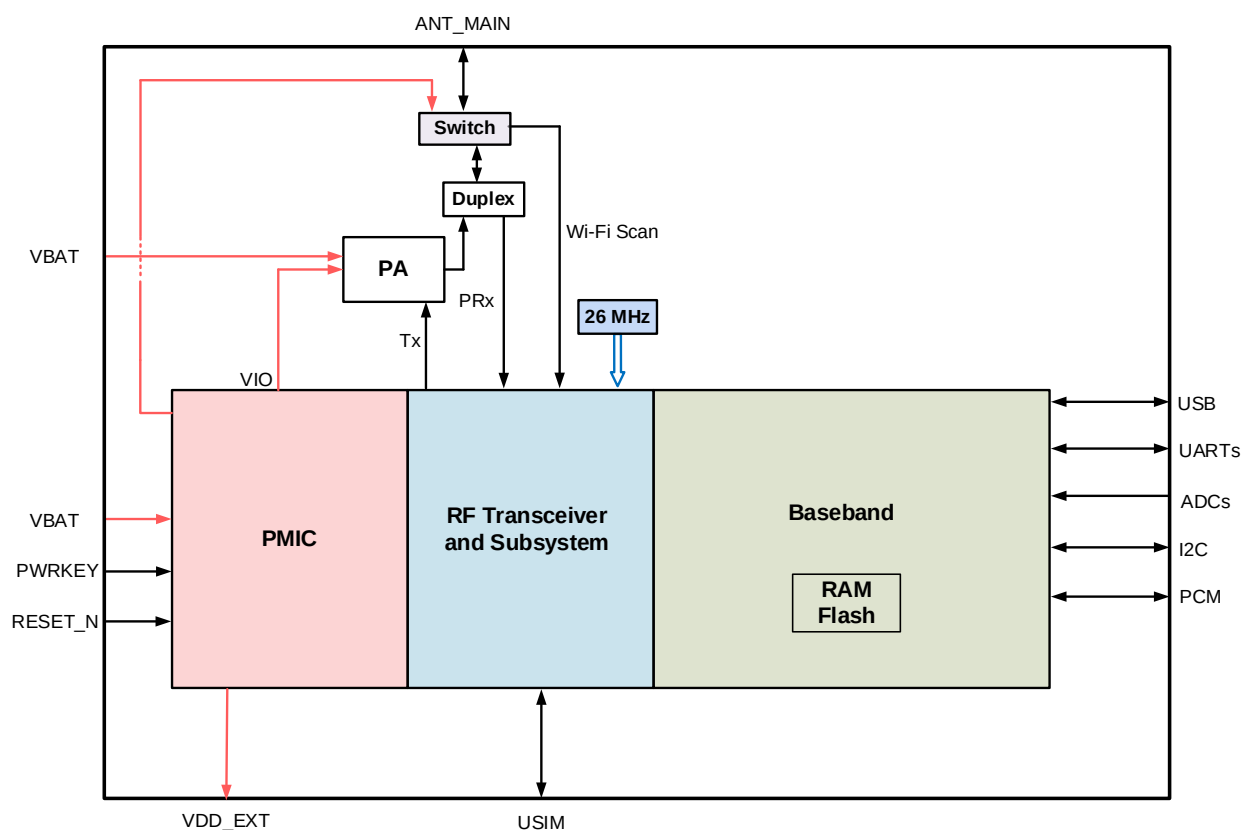


Figure 1: Block Diagram

2.4. Pin Assignment

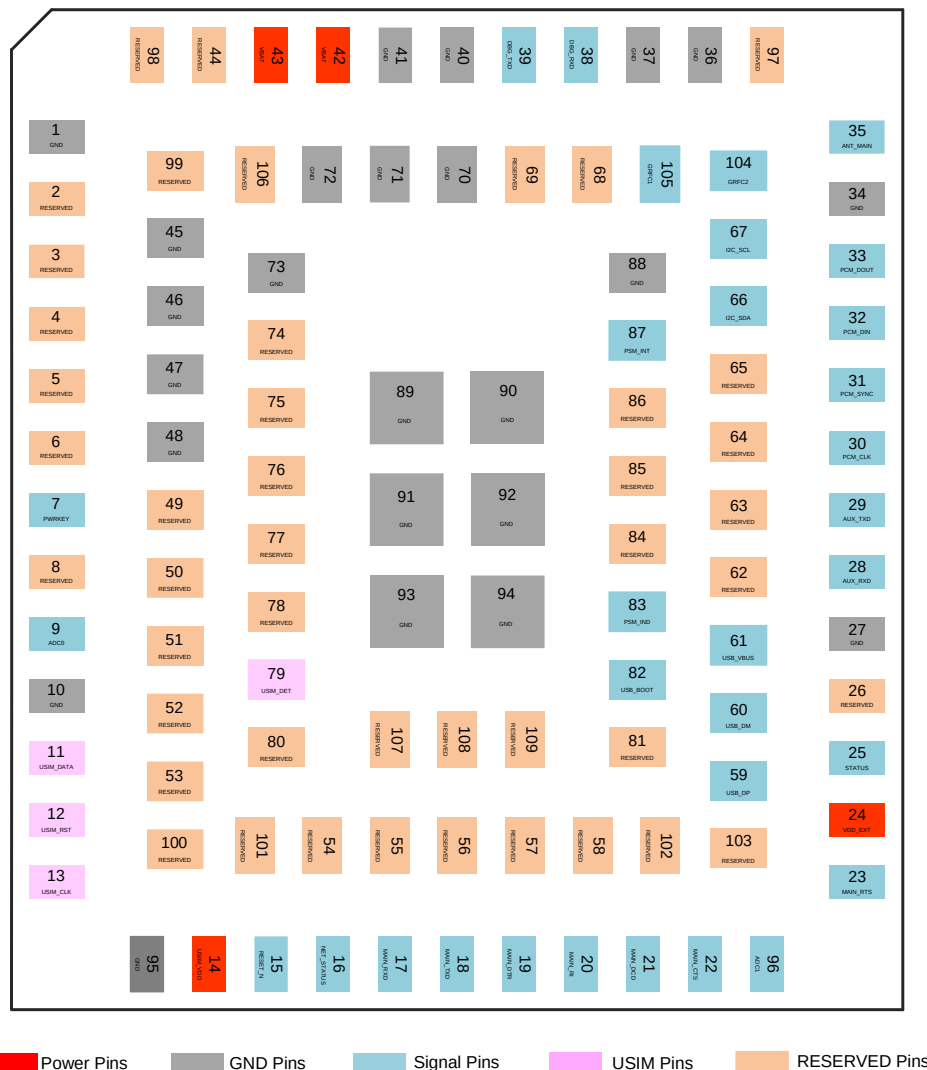


Figure 2: Pin Assignment (Top View)

NOTE

- Do the following only if the module should be put into forced download mode: pull USB_BOOT (pin 82) to the ground before the module successfully starts up.
- In sleep mode, some pins of the main UART interface (pins 17, 18, 22, 23), auxiliary UART interface* (pins 28, 29), debug UART interface (pins 38, 39), USB_BOOT (pin 82), PCM* and I2C* interfaces (pins 30–33, 66, 67), GRFC interfaces (pins 104, 105) are powered down, and the high-level pins will output periodic pulses with the paging cycle. The driving capacity, status indication and data transmission functions of these pins will be invalid. Take this into consideration when designing circuits.

3. Keep all RESERVED pins and unused pins unconnected.
4. Ensure that the pull-up power supply of the module's pins is VDD_EXT or controlled by VDD_EXT, and there is no current sink on the module's pins before the module turns on. For more details, contact Quectel Technical Support.

2.5. Pin Description

The following table shows the pin descriptions.

Table 5: I/O Parameters Definition

Type	Description
AI	Analog Input
AIO	Analog Input/Output
DI	Digital Input
DO	Digital Output
DIO	Digital Input/Output
PI	Power Input
PO	Power Output
OD	Open Drain

DC characteristics include power domain and rated current.

Table 6: Pin Description

Power Supply					
Pin Name	Pin No.	I/O	Description	DC Characteristics	Comment
VBAT	42, 43	PI	Power supply for the module	Vmax = 4.3 V Vmin = 3.3 V Vnom = 3.8 V	External power supply is recommended to provide with sufficient current of 1.5 A. It is recommended to add an external TVS.

					A test point is recommended to be reserved.
VDD_EXT	24	PO	Provide 1.8 V for external circuit	Vnom = 1.8 V I _O max = 50 mA	A test point is recommended to be reserved.
GND	1, 10, 27, 34, 36, 37, 40, 41, 45–48, 70–73, 88–95				

Turn On/Off

Pin Name	Pin No.	I/O	Description	DC Characteristics	Comment
PWRKEY	7	DI	Turn on/off the module	V _{IL} max = 0.5 V	Active low. A test point is recommended to be reserved.
RESET_N	15	DI	Reset the module		Active low. A test point is recommended to be reserved if unused.

Indication Signals

Pin Name	Pin No.	I/O	Description	DC Characteristics	Comment
STATUS	25	DO	Indicate the module's operation status	V _{OH} min = 1.44 V V _{OL} max = 0.27 V	If unused, keep them open.
NET_STATUS	16	DO	Indicate the module's network activity status		

USB Interface

Pin Name	Pin No.	I/O	Description	DC Characteristics	Comment
USB_VBUS	61	AI	USB connection detect	Vmax = 5.25 V Vmin = 3.0 V Vnom = 5.0 V	A test point must be reserved.
USB_DP	59	AIO	USB differential data (+)		USB 2.0 compliant. Requires differential impedance of 90 Ω. Test points must be reserved.
USB_DM	60	AIO	USB differential data (-)		

USIM Interface

Pin Name	Pin	I/O	Description	DC	Comment
----------	-----	-----	-------------	----	---------

No.				Characteristics	
				I _o max = 37 mA	
USIM_VDD	14	PO	USIM card power supply	Low-voltage: V _{max} = 1.85 V V _{min} = 1.75 V High-voltage: V _{max} = 3.05 V V _{min} = 2.95 V	Either 1.8 V or 3.0 V USIM card is supported and can be identified automatically by the module.
USIM_DATA	11	DIO	USIM card data	USIM_VDD	
USIM_CLK	13	DO	USIM card clock		
USIM_RST	12	DO	USIM card reset		
USIM_DET	79	DI	USIM card hot-plug detect	V _{IH} min = 1.33 V V _{IL} max = 0.42 V	If unused, keep it open.
Auxiliary UART Interface*					
Pin Name	Pin No.	I/O	Description	DC Characteristics	Comment
AUX_TXD	29	DO	Auxiliary UART transmit	VDD_EXT	If unused, keep them open.
AUX_RXD	28	DI	Auxiliary UART receive		
Main UART Interface					
Pin Name	Pin No.	I/O	Description	DC Characteristics	Comment
MAIN_CTS	22	DO	Clear to send signal from the module	VDD_EXT	Connect to MCU's CTS. If unused, keep it open.
MAIN_RTS	23	DI	Request to send signal to the module		Connect to MCU's RTS. If unused, keep it open.
MAIN_RXD	17	DI	Main UART receive		If unused, keep them open.
MAIN_TXD	18	DO	Main UART transmit		
MAIN_DCD	21	DO	Main UART data carrier detect		
MAIN_RI	20	DO	Main UART ring indication	V _{OH} min = 1.44 V V _{OL} max = 0.27 V	If unused, keep them open.
MAIN_DTR	19	DI	Main UART data terminal ready	V _{IH} min = 1.33 V V _{IL} max = 0.42 V	
Debug UART Interface					

Pin Name	Pin No.	I/O	Description	DC Characteristics	Comment
DBG_RXD	38	DI	Debug UART receive	VDD_EXT	Test points must be reserved.
DBG_TXD	39	DO	Debug UART transmit		
I2C Interface*					
Pin Name	Pin No.	I/O	Description	DC Characteristics	Comment
I2C_SCL	67	OD	I2C serial clock	VDD_EXT	External pull-up resistor is required. If unused, keep them open.
I2C_SDA	66	OD	I2C serial data		
PCM Interface*					
Pin Name	Pin No.	I/O	Description	DC Characteristics	Comment
PCM_SYNC	31	DO	PCM data frame sync	VDD_EXT	If unused, keep them open.
PCM_CLK	30	DO	PCM clock		
PCM_DIN	32	DI	PCM data input		
PCM_DOUT	33	DO	PCM data output		
RF Antenna Interface					
Pin Name	Pin No.	I/O	Description	DC Characteristics	Comment
ANT_MAIN ³	35	AIO	Main antenna/ Wi-Fi Scan antenna interface		50 Ω impedance.
ADC Interfaces					
Pin Name	Pin No.	I/O	Description	DC Characteristics	Comment
ADC0	9	AI	General-purpose ADC interface	Input voltage range: 0–1.2 V	If unused, keep them open.
ADC1	96	AI	General-purpose ADC interface		
Other Interfaces					
Pin Name	Pin No.	I/O	Description	DC Characteristics	Comment

³ Main antenna and Wi-Fi Scan antenna only support passive antennas.

USB_BOOT	82	DI	Force the module into download mode	VDD_EXT	Active low. A test point must be reserved.
PSM_IND*	83	DO	Indicate the module's power saving mode	V _{OH} min = 1.44 V V _{OL} max = 0.27 V	If unused, keep it open.
PSM_INT*	87	DI	External interrupt; wake up the module from power saving mode	V _{IH} min = 1.33 V V _{IL} max = 0.42 V	Externally pulling up this pin can make the module exit power saving mode. If unused, keep it open.
GRFC2	104	DO	Generic RF controller		If unused, keep them open.
GRFC1	105	DO	Generic RF controller		
RESERVED	2–6, 8, 26, 44, 49–58, 62–65, 68, 69, 74–78, 80, 81, 84–86, 97–103, 106-109				Keep these pins open.

Some pins of the module are divided into three types: WAKEUP, AGPIO, and AGPIOWU pins. The pin characteristics of these three types are as follows.

Table 7: WAKEUP & AGPIO & AGPIOWU Pin Characteristics

WAKEUP Pins	Characteristics
- PSM_INT* - USB_VBUS - USIM_DET	- Support wake-up interrupt function. - High-level voltage: about 1.2 V for PSM_INT and USIM_DET. - The state of the pins will not be affected even if the module enters the sleep mode.
AGPIO Pins	Characteristics
- MAIN_DCD - PSM_IND* - STATUS - NET_STATUS - MAIN_RI	The state of the pins will not be affected even if the module enters the sleep mode.
AGPIOWU Pin	Characteristics
MAIN_DTR	- Supports wake-up interrupt function. - High-level voltage: about 1.2 V. - The state of the pins will not be affected even if the module enters the sleep mode.

2.6. EVB Kit

Quectel supplies an evaluation board (UMTS<E EVB) with accessories to develop and test the module. For more details, see **document [1]**.

3 Operating Characteristics

3.1. Operating Modes

Table 8: Overview of Operating Modes

Mode	Details	
Full Functionality Mode	Idle	Software is active. The module is registered on the network but has no data interaction with the network.
	Data	Network connection is ongoing. In this mode, the power consumption is decided by network setting and data transfer rate.
Minimum Functionality Mode	● Executing AT+CFUN =0 sets the module to minimum functionality mode. ● Both RF function and USIM card are invalid.	
Airplane Mode	● Executing AT+CFUN =4 sets the module to airplane mode. ● RF function is invalid.	
Sleep Mode	Power consumption of the module is reduced to the minimal level, but the module can still receive paging, SMS and TCP/UDP data from the network.	
eDRX	The module and network can negotiate eDRX parameters through Non-Access Stratum (NAS) signalling to reduce power consumption. In this mode, downlink data transmission or signalling process (such as paging) initiated by the network should wait until the terminal enters the preset Paging Time Window (PTW), and the resulting communication delay is determined by the length of the eDRX cycle.	
Turn-off Mode	The VBAT power supply for the module remains applied, and software is not executed.	

NOTE

For more details about **AT+CFUN**, see *document [2]*.

3.2. Sleep Mode

The power consumption of the module is reduced to an ultra-low level during sleep mode.

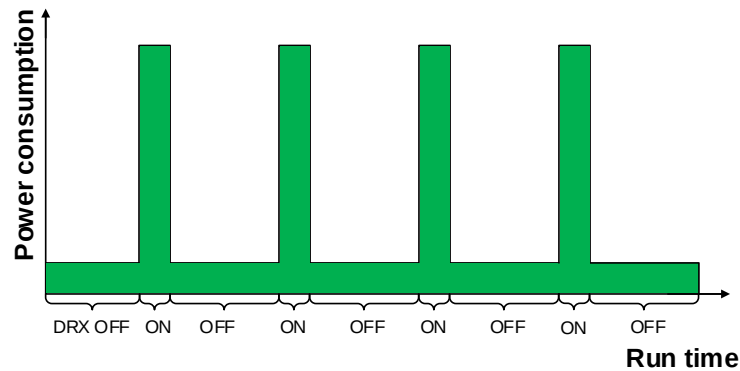


Figure 3: Power Consumption During Sleep Mode

NOTE

1. The DRX period value is sent by the base station over the wireless network.
2. In sleep mode, some pins of the main UART interface (pins 17, 18, 22, 23), auxiliary UART interface* (pins 28, 29), debug UART interface (pins 38, 39), USB_BOOT (pin 82), PCM* and I2C* interfaces (pins 30–33, 66, 67), GRFC interfaces (pins 104, 105) are powered down, and the high-level pins will output periodic pulses with the paging cycle. The driving capacity, status indication and data transmission functions of these pins will be invalid. Take this into consideration when designing circuits.

3.2.1. UART Application Scenario

If the MCU communicates with the module via UART interface, the following two preconditions should be met to make the module enter sleep mode.

- Execute **AT+QSCCLK=1**. For more details, see **document [2]**.
- Ensure MAIN_DTR is held high or is kept unconnected.

The figure illustrates the connection between the module and the MCU.

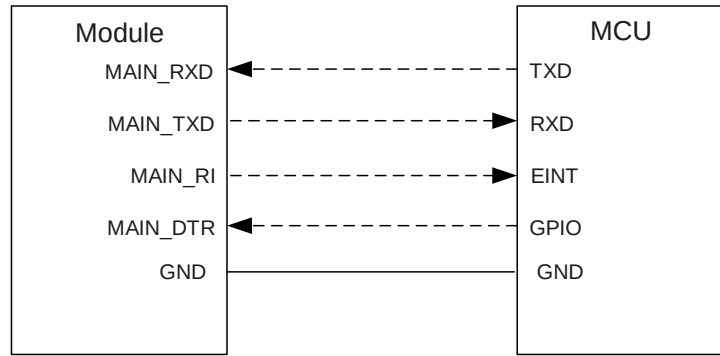


Figure 4: Sleep Mode Application via UART

- Driving the MAIN_DTR low will wake up the module.
- When the module has a URC to report, MAIN_RI signal will wake up the MCU. See **Chapter 4.8.3** for details about MAIN_RI behavior.

NOTE

Pay attention to the level match shown in dotted line between the module and the MCU.

3.2.2. USB Application Scenario

For the two situations below, three preconditions must be met to make the module enter sleep mode.

- Execute **AT+QSCCLK=1**.
- Ensure the MAIN_DTR is held high or is kept unconnected.
- Ensure the host's USB bus, which is connected with the module's USB interface, enters suspend state.

3.2.2.1. USB Application with USB Suspend/Resume & Remote Wakeup Function

The host supports USB suspend, resume and remote wakeup function. The figure illustrates the connection between the module and the host.

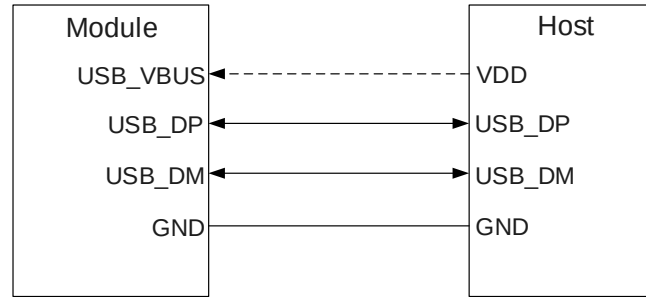


Figure 5: Sleep Mode Application with USB Remote Wakeup

- Sending data to the module through USB will wake up the module.
- When the module has a URC to report, the module will send remote wake-up signals through USB bus to wake up the host.

3.2.2.2. USB Application with USB Suspend/Resume and MAIN_RI Function

If the host supports USB suspend and resume, but does not support remote wakeup function, the MAIN_RI signal is needed to wake up the host. The following figure illustrates the connection between the module and the host.

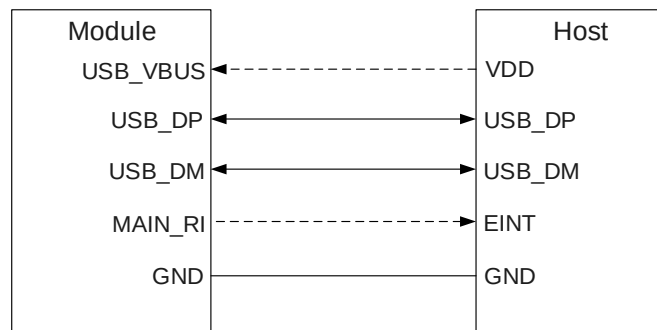


Figure 6: Sleep Mode Application with MAIN_RI

- Sending data to the module through USB will wake up the module.
- When the module has a URC to report, MAIN_RI signal will wake up the host. See **Chapter 4.8.3** for details about MAIN_RI behavior.

3.2.2.3. USB Application without USB Suspend Function

If the host does not support USB suspend function, the following three preconditions should be met to make the module enter sleep mode.

- Execute **AT+QSClk=1**.
- Ensure the MAIN_DTR is held high or is kept unconnected.
- Disconnect the USB_VBUS power supply.

If the host does not support USB suspend function, USB_VBUS should be disconnected through an external control circuit to make the module enter sleep mode. The figure illustrates the connection between the module and the host.

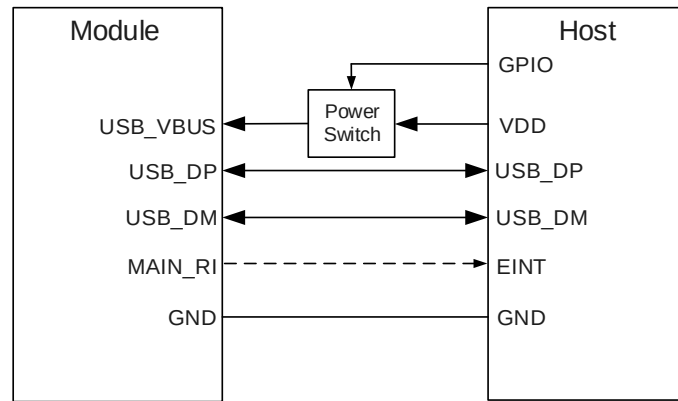


Figure 7: Sleep Mode Application without USB Suspend

Restore the supply power of USB_VBUS will wake up the module.

NOTE

Pay attention to the level match shown in dotted line between the module and the host.

3.3. eDRX

The module supports eDRX mode. When the module is working normally, use **AT+CEDRXS** to enter eDRX mode and **AT+QSClk=1** to enter sleep mode. The module can be awakened from eDRX by MAIN_DTR pin.

To minimize power consumption in eDRX, VDD_EXT should be disabled with **AT+QCFG="sleep/aon_cfg"**. Consequently, for pins with wakeup functions such as MAIN_DTR,

USB_VBUS, MAIN_RXD, PSM_INT* and USIM_DET, if VDD_EXT serves as the pull-up power supply, it must be replaced with an external power source. Failure to do this may lead to abnormalities in sleep mode.

NOTE

1. For details about AT commands, please refer to **document [2]** and **[3]**.
2. Ensure the network supports eDRX.

3.4. Airplane Mode

When the module enters airplane mode, the RF function is disabled and all related AT commands cannot be executed. This mode can be set through **AT+CFUN=<fun>**, which allows you to choose the functionality level through setting <fun> as **0**, **1** or **4**. For more details, see **document [2]**.

- **AT+CFUN=0**: Minimum functionality mode. (Both RF and USIM functions are disabled)
- **AT+CFUN=1**: Full functionality mode. (Default).
- **AT+CFUN=4**: Airplane mode. (RF function is disabled)

3.5. Power Supply

3.5.1. Power Supply Pins

The module has two VBAT pins for connecting to external power supply.

Table 9: Pin Definition of Power Supply

Pin Name	Pin No.	I/O	Description	Min.	Typ.	Max.	Unit
VBAT	42, 43	PI	Power supply for the module	3.3	3.8	4.3	V
GND	1, 10, 27, 34, 36, 37, 40, 41, 45–48, 70–73, 88–95						

3.5.2. Power Supply Reference Design

Power supply design is essential for module performance. It is recommended to be provided with sufficient current of 1.5 A to the module. If the voltage difference between input voltage and the supply voltage is small, it is suggested to use an LDO; if the voltage difference is big, a buck converter is recommended.

A reference design for +5 V input power source is illustrated in the following figure (Please adjust the parameters in accordance with the actual situation).

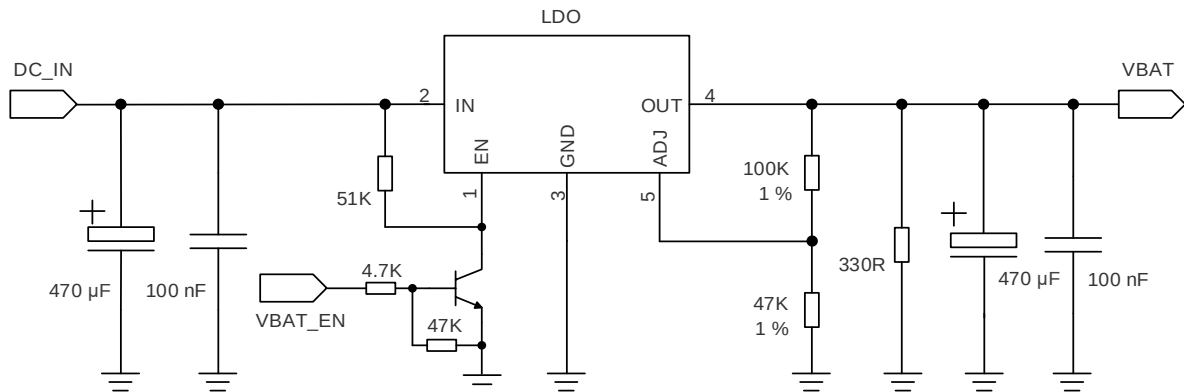


Figure 8: Power Supply Reference Design

3.5.3. Voltage Stability Requirements

The power supply range of the module is 3.3–4.3 V. Make sure the input voltage never drops below 3.3 V.

To decrease the voltage drop, use a bypass capacitor of about 100 µF with low ESR ($ESR \leq 0.7 \Omega$). A multi-layer ceramic chip (MLCC) capacitor array with ultra-low ESR is recommended. Use three ceramic capacitors (100 nF, 10 pF and 33 pF) to compose the array and place them close to VBAT pins. If the module is powered from a single voltage source, then the power supply distribution path can be traced with a star topology structure when connect to the module. The width of VBAT trace should be at least 1.5 mm. In principle, the longer the VBAT trace is, the wider it should be.

To avoid power ripples and surges and ensure the stability of the power supply, add a TVS with $V_{RWM} = 4.7$ V, low clamping voltage and high reverse peak pulse current I_{PP} at the front end of the power supply. If the TVS is not added, surges on the input power supplies may cause damage to the module. Reference design is shown below.

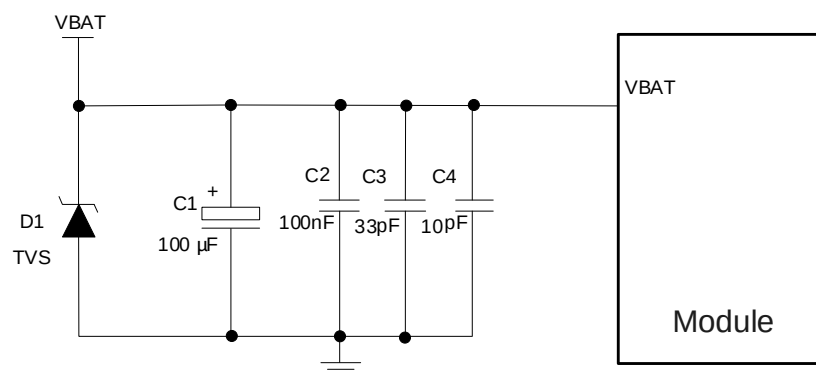


Figure 9: Power Supply Reference Design

3.6. Turn On

3.6.1. Turn On with PWRKEY

Table 10: Pin Definition of PWRKEY

Pin Name	Pin No.	I/O	Description	Comment
PWRKEY	7	DI	Turn on/off the module	Active low. A test point is recommended to be reserved.

When the module is in power-down mode, it can be turned on by driving the PWRKEY low for **at least 500 ms**. It is recommended to use an open collector driver to control the PWRKEY.

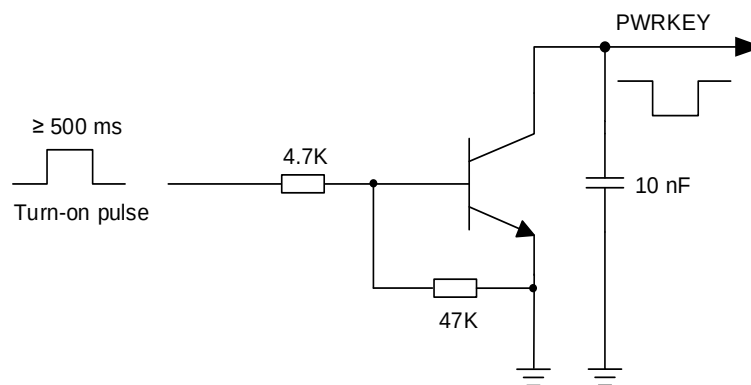


Figure 10: Reference Design of Turning on the Module with Driving Circuit

The module can also be turned on by pressing the PWRKEY button. A TVS diode should be placed near the button for protection against ESD.

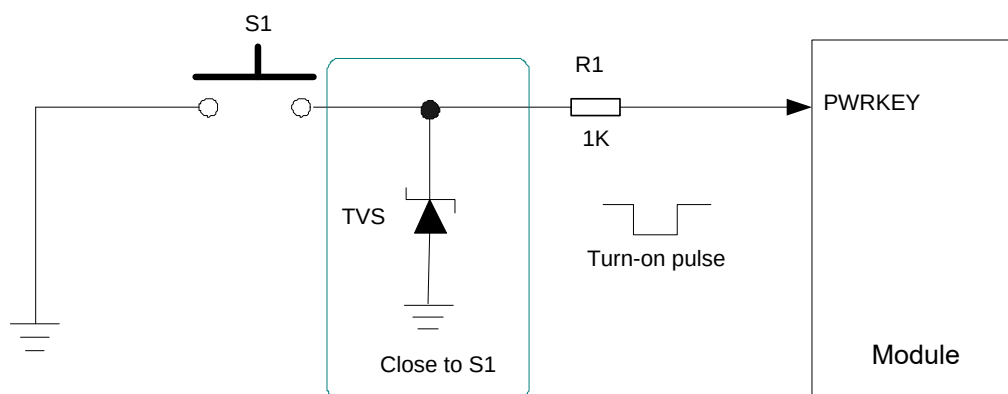


Figure 11: Reference Design of Turning on Module with Keystroke

The turn-on timing is illustrated in the following figure.

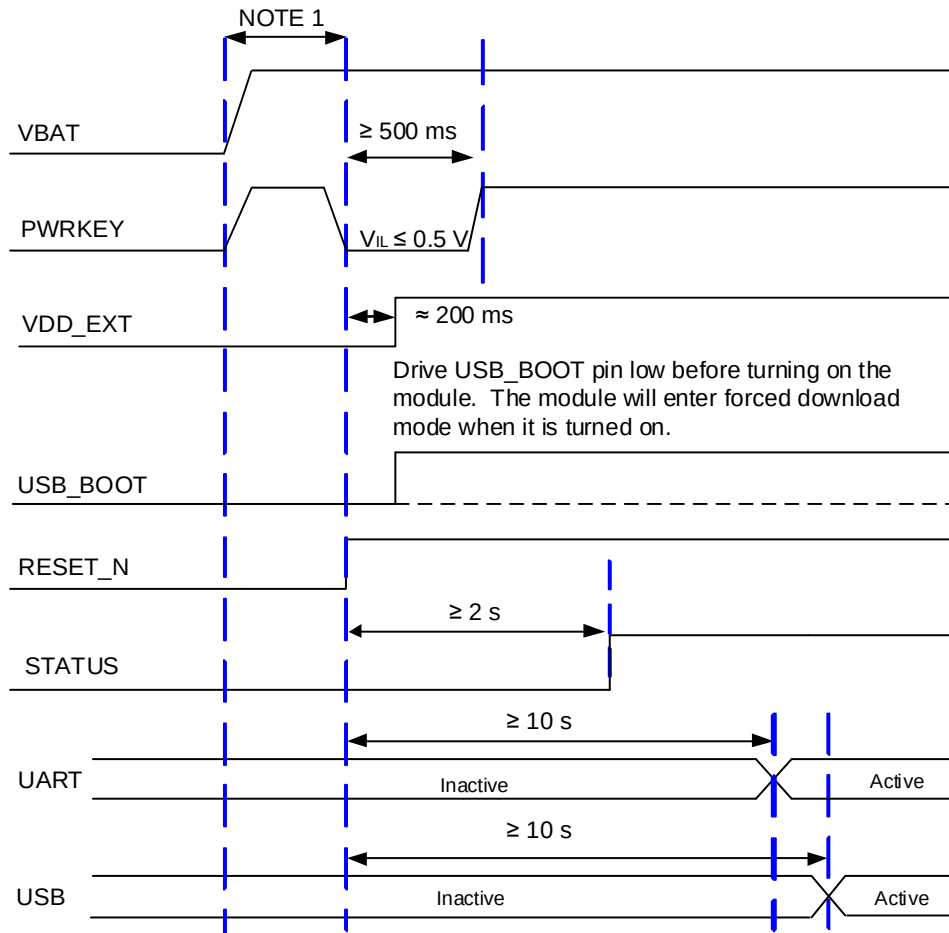


Figure 12: Turn-on Timing

NOTE

1. Ensure a stable VBAT for at least 30 ms before driving the PWRKEY low.
2. If the module needs to turn on automatically but does not need turn-off function, you can drive PWRKEY pin low by connecting it directly to the ground with a recommended 4.7 k Ω resistor or by using a GPIO to drive PWRKEY low before the module turns on (PWRKEY needs to remain at low level after the module is turned on).

3.7. Turn Off

The following procedures can be used to turn off the module normally.

3.7.1. Turn Off with PWRKEY

The module will execute turn-off procedure if you drive the PWRKEY pin low for **at least 650 ms** and then release it.

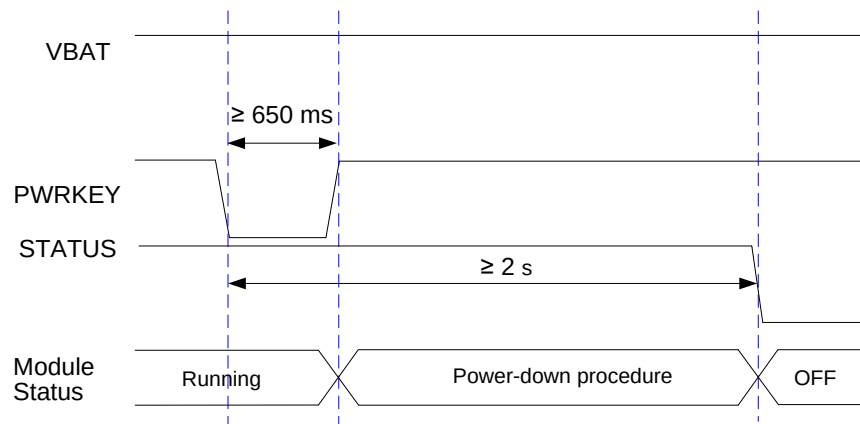


Figure 13: Timing of Turn-off with PWRKEY

3.7.2. Turn Off with AT Command

You can execute **AT+QPOWD** to turn off the module, which is the same as turning off the module via the PWRKEY pin. See **document [2]** for details about **AT+QPOWD**.

NOTE

1. When PWRKEY is pulled down to the ground, **AT+QPOWD** cannot be used to turn off the module.
2. When turning off the module with the AT command, keep PWRKEY at high level after the execution of the command. Otherwise, the module will be turned on automatically again after successful turn-off.

3.8. Reset

Pulling down PWRKEY when RESET_N is at low level can reset the module. The RESET_N signal is sensitive to interference. Therefore, it is recommended to route the trace as short as possible and surround it with ground.

Table 11: Pin Definition of RESET_N

Pin Name	Pin No.	I/O	Description	Comment
RESET_N	15	DI	Reset the module	Active low. A test point is recommended to be reserved if unused.

You can use an open collector driver to control RESET_N and PWRKEY pins.

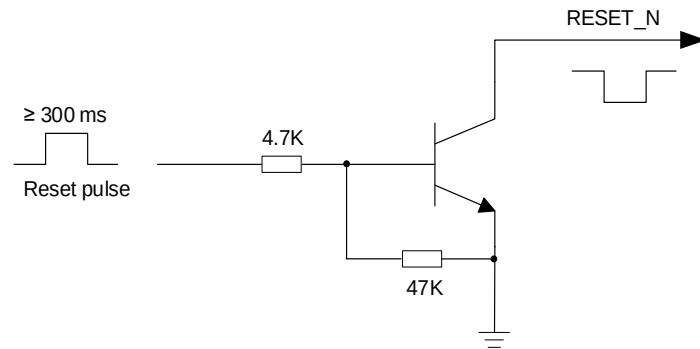


Figure 14: Reference Design of RESET_N with Driving Circuit

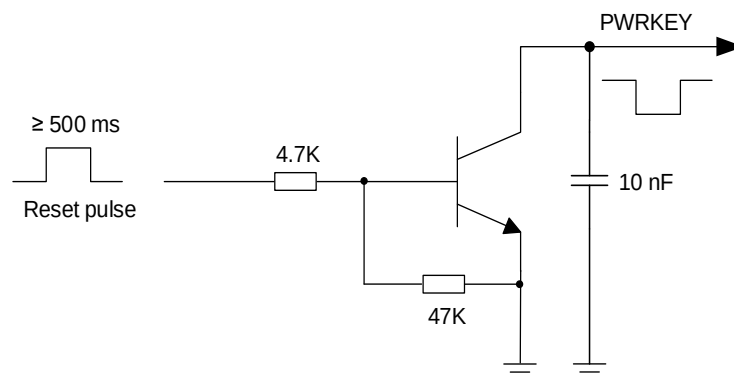


Figure 15: Reference Design of PWRKEY with Driving Circuit

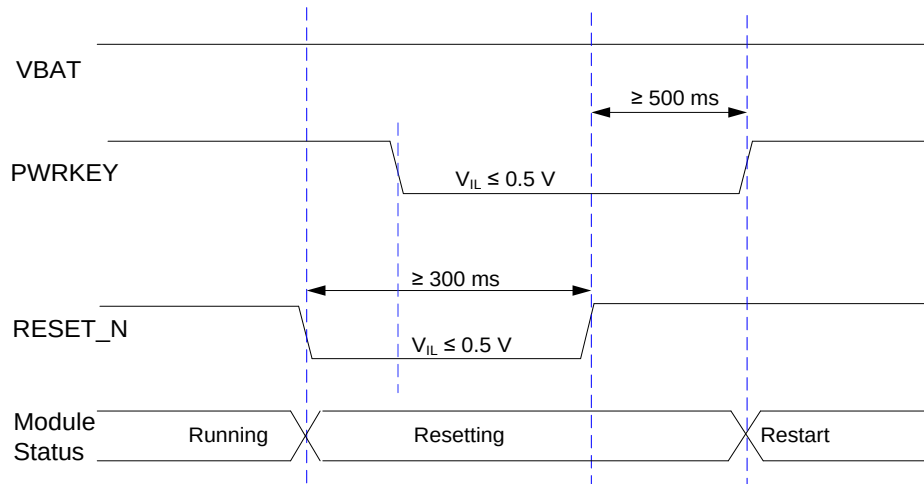


Figure 16: Reset Timing

NOTE

1. Pull down PWRKEY when RESET_N is at low level.
2. Ensure that the capacitances connected to PWRKEY and RESET_N do not exceed 10 nF.
3. When PWRKEY is pulled down to the ground, the module can be reset by pulling RESET_N to low level.

4 Application Interfaces

4.1. USB Interface

The module has one USB interface, which complies with the USB 2.0 specifications, and supports high speed (480 Mbps) and full speed (12 Mbps) on USB 2.0. The module only supports USB slave mode. The USB interface can be used for AT command communication, data transmission, software debugging, firmware upgrading and outputting logs.

Table 12: Pin Definition of USB Interface

Pin Name	Pin No.	I/O	Description	Comment
USB_VBUS	61	AI	USB connection detect	A test point must be reserved.
USB_DP	59	AIO	USB differential data (+)	USB 2.0 compliant. Require differential impedance of 90 Ω .
USB_DM	60	AIO	USB differential data (-)	Test points must be reserved.

It is recommended to use USB interface for firmware upgrading. Test points must be reserved so that logs can be obtained and customers' issues can be found.

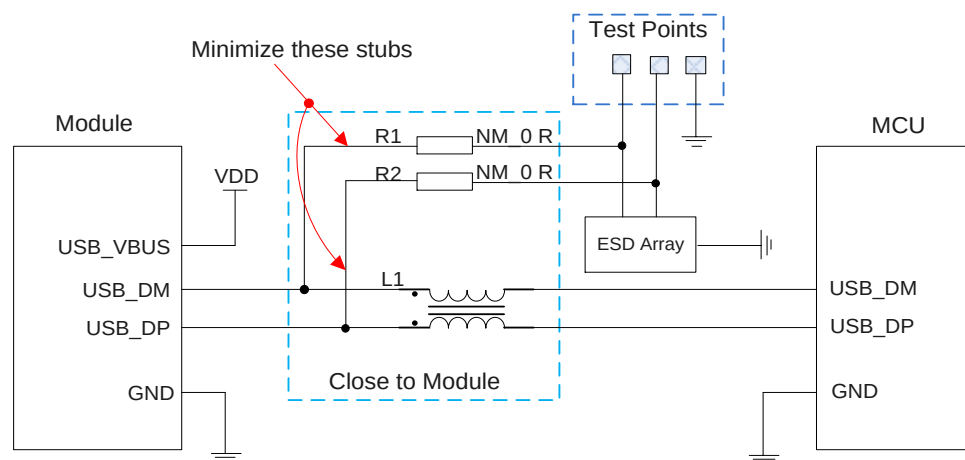


Figure 17: Reference Circuit of USB Interface

It is recommended to add a common-mode choke L1 in series between the module and MCU to suppress EMI. In addition, the 0 Ω resistors (R1 and R2) should be added in series between the module and the test points for debugging. These resistors are not mounted by default. To ensure USB data transmission integrity, L1, R1 and R2 must be placed close to the module, and resistors R1 and R2 should be placed close to each other. Extra trace stubs must be as short as possible.

To ensure performance, you should follow the following principles when designing a USB interface:

- Route the USB signal traces as differential pairs surrounded by ground. The impedance of USB 2.0 differential trace is 90 Ω .
- Do not route signal traces under crystals, oscillators, magnetic devices and RF signal traces. It is important to route the USB differential traces in an inner layer of the PCB, and surround the traces with ground on the same layer and with ground planes above and below.
- Pay attention to the impact caused by stray capacitance of the ESD protection component on USB data traces. Typically, the stray capacitance should be less than 2 pF.
- Place ESD protection components as close to the USB interface as possible.

For more details about the USB specifications, visit <http://www.usb.org/home>.

4.2. USB_BOOT

The module has a USB_BOOT pin. If you drive USB_BOOT to the ground before turning on the module, the module will enter forced download mode when turned on. In this mode, the module supports firmware upgrade over USB 2.0 interface.

Table 13: Pin Definition of USB_BOOT

Pin Name	Pin No.	I/O	Description	Comment
USB_BOOT	82	DI	Force the module into download mode	Active low. A test point must be reserved.

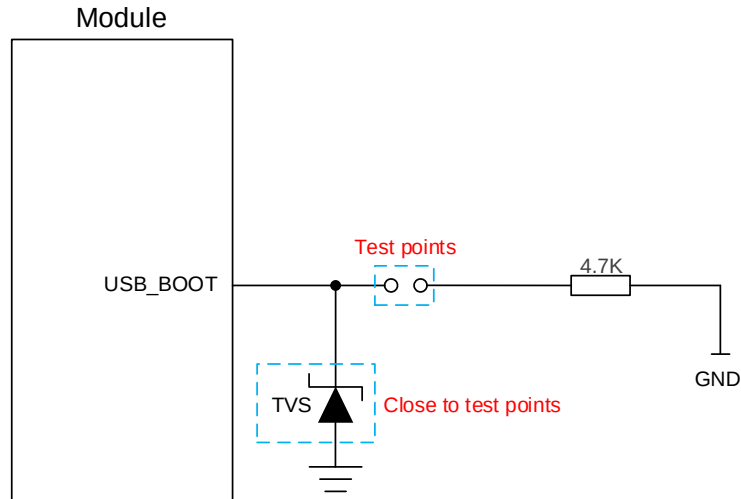


Figure 18: Reference Design of USB_BOOT Interface

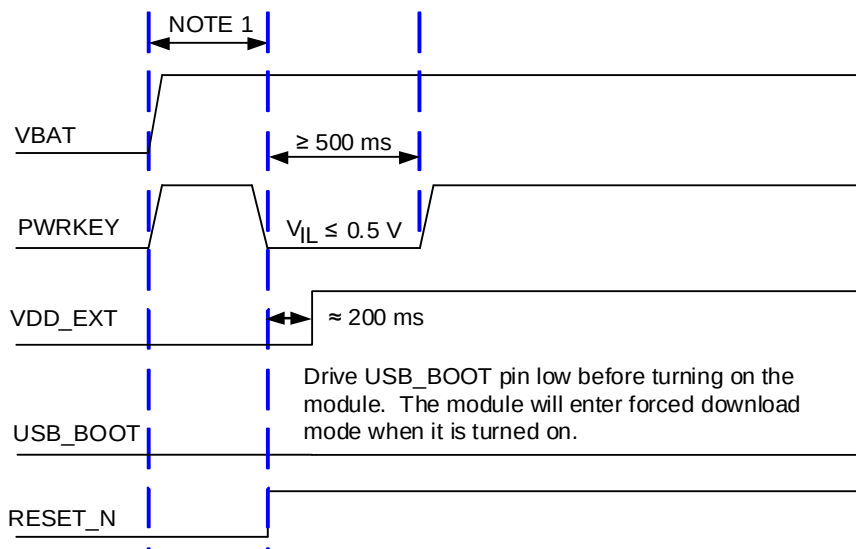


Figure 19: Timing of Entering Forced Download Mode

NOTE

1. Ensure a stable VBAT for at least 30 ms before driving the PWRKEY low.
2. When using MCU to put the module to forced download mode, follow the above timing. If you need to manually force the module to enter forced download mode, connect the test points as shown in **Figure 18**.

4.3. UART Interfaces

The module has 3 UART interfaces.

Table 14: UART Information

UART Types	Supported Baud Rates (bps)	Default Baud Rates (bps)	Functions
Main UART interface	4800, 9600, 19200, 38400, 57600, 115200, 230400, 460800, 921600	115200	● AT command communication and data transmission ● RTS and CTS hardware flow control
Debug UART interface	115200, 3000000	3000000	● Partial log output
Auxiliary UART interface*	115200	115200	-

Table 15: Pin Definition of UART Interfaces

Pin Name	Pin No.	I/O	Description	Comment
MAIN_CTS	22	DO	Clear to send signal from the module	Connect to MCU's CTS. If unused, keep it open.
MAIN_RTS	23	DI	Request to send signal to the module	Connect to MCU's RTS. If unused, keep it open.
MAIN_RXD	17	DI	Main UART receive	If unused, keep them open.
MAIN_DCD	21	DO	Main UART data carrier detect	
MAIN_TXD	18	DO	Main UART transmit	
MAIN_RI	20	DO	Main UART ring indication	
MAIN_DTR	19	DI	Main UART data terminal ready	
AUX_TXD*	29	DO	Auxiliary UART transmit.	
AUX_RXD*	28	DI	Auxiliary UART receive.	
DBG_RXD	38	DI	Debug UART transmit	
DBG_TXD	39	DO	Debug UART receive	Test points must be reserved.

The module has **1.8 V UART interfaces**. If the level of the MCU is 1.8 V, and the MAIN_TXD of the module is connected to the RXD of the MCU, the MAIN_TXD must be connected to a 10 kΩ resistor and pulled up to 1.8 V to prevent the MCU from receiving error messages when the module is in sleep mode. A voltage-level translator should be used if the application features a 3.3 V UART interface.

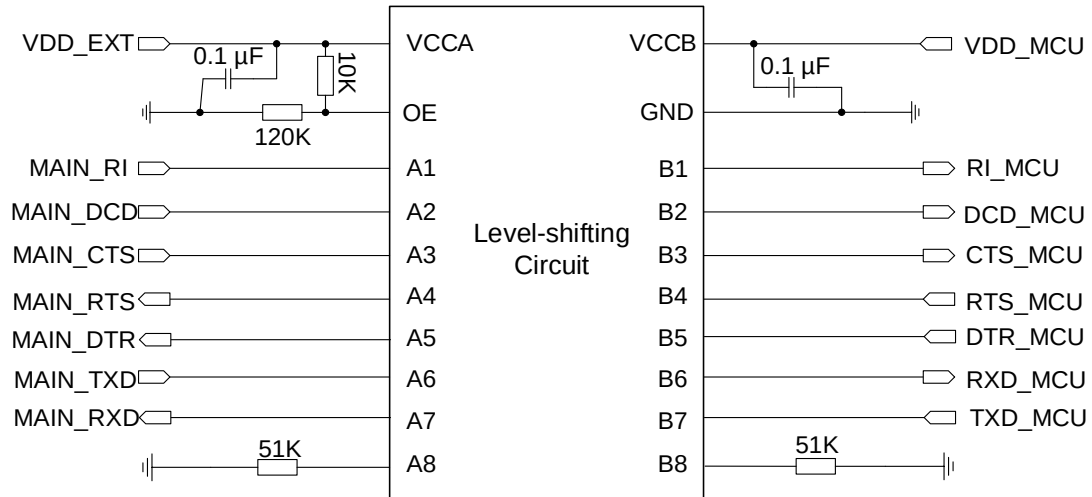


Figure 20: Reference Design of UART with Voltage-Level Translator (Main UART)

Another example of level-shifting circuit is shown below. For the design of input/output circuits in dotted lines, see the ones in solid lines, but pay attention to the direction of the connection.

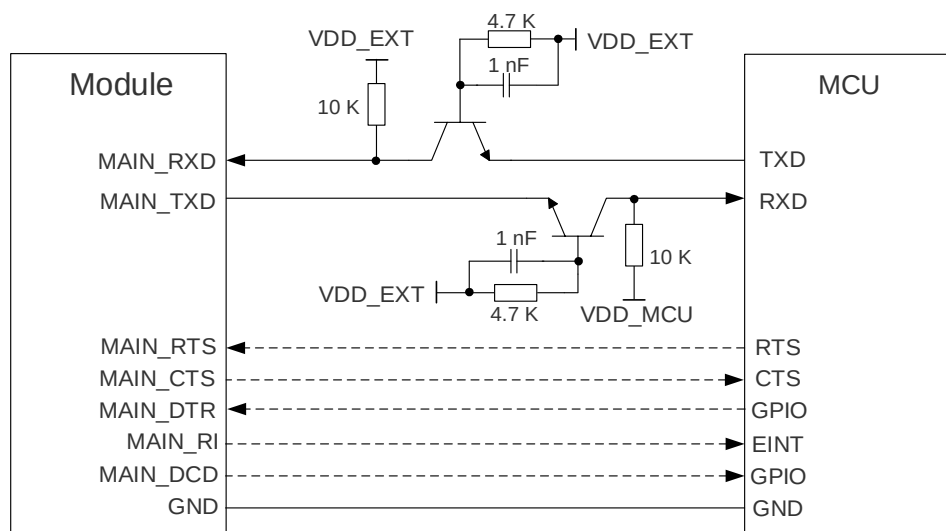


Figure 21: Reference Design of UART with Transistor Level-Shifting Circuit (Main UART)

NOTE

1. Transistor level-shifting circuit above is not suitable for applications with baud rates exceeding 460 kbps.
2. Please note that the module's CTS is connected to MCU's CTS, and the module's RTS is connected to MCU's RTS.
3. The level-shifting circuits (**Figure 20** and **Figure 21**) take the main UART as an example. The circuits of the debug UART and the auxiliary UART are connected in the same way as the main UART.
4. To increase the stability of UART communication, it is recommended to add UART hardware flow control design.

4.4. USIM Interface

The USIM interface meets ETSI and IMT-2000 requirements. Both the 1.8 V and the 3.0 V USIM cards are supported.

Table 16: Pin Definition of USIM Interface

Pin Name	Pin No.	I/O	Description	Comment
USIM_VDD	14	PO	USIM card power supply	Either 1.8 V or 3.0 V USIM card is supported and can be identified automatically by the module.
USIM_DATA	11	DIO	USIM card data	
USIM_CLK	13	DO	USIM card clock	
USIM_RST	12	DO	USIM card reset	
USIM_DET	79	DI	USIM card hot-plug detect	If unused, keep it open.

The module supports USIM card hot-plug detection via the USIM_DET pin, and both high and low level detection are supported. The function is disabled by default, and see **AT+QSIMDET** in **document [2]** for more details.

A reference design of USIM card interface with an 8-pin USIM card connector is illustrated in the following figure.

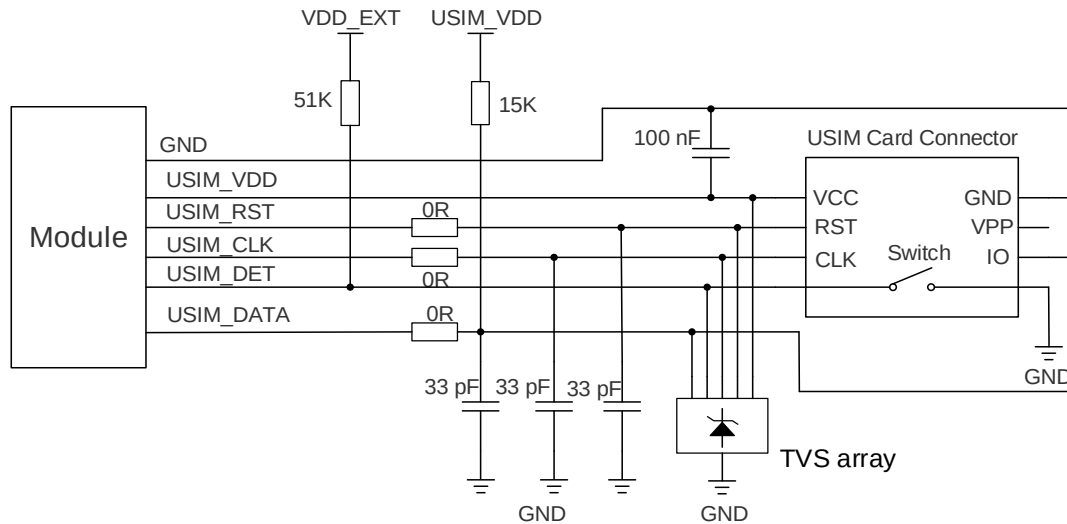


Figure 22: Reference Design of USIM Interface with an 8-Pin USIM Card Connector

If USIM card detection function is not needed, keep USIM_DET disconnected. A reference design of USIM interface with a 6-pin USIM card connector is illustrated in the following figure.

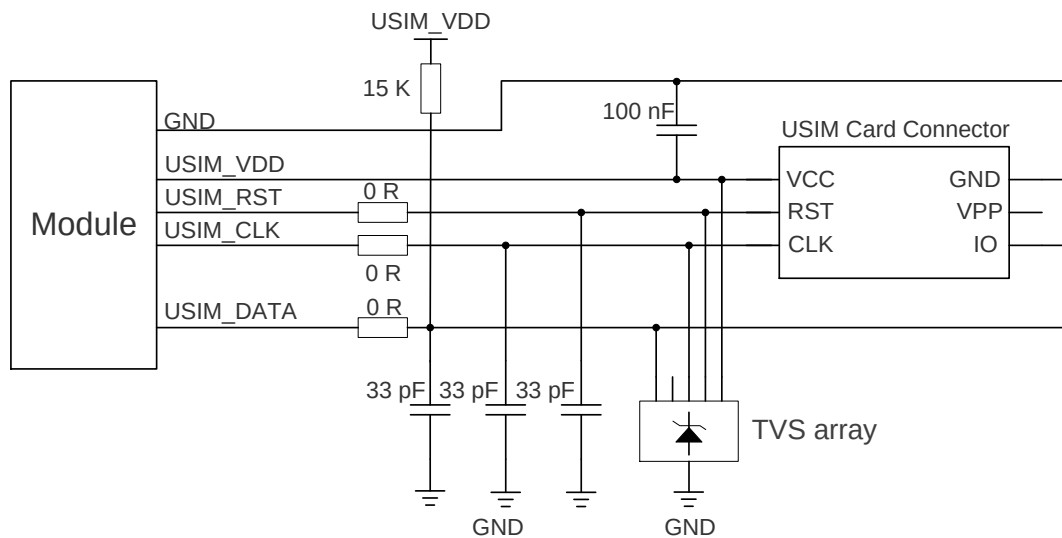


Figure 23: Reference Design of USIM Interface with a 6-Pin USIM Card Connector

To enhance USIM card reliability and availability in applications, follow the principles below in the USIM circuit design:

- Place the USIM card connector close to the module. Keep the trace length as short as possible, at most 200 mm.
- Keep USIM card signal traces away from RF and power supply traces.
- Make sure that the bypass capacitor between USIM_VDD and GND does not exceed 1 μ F, and should be placed close to the USIM card connector.

- To avoid cross talk between USIM_DATA and USIM_CLK, keep the traces away from each other and shield them by surrounding them with ground.
- To improve ESD protection, it is recommended to add a TVS array on USIM pins. The parasitic capacitance of the TVS array should not exceed 15 pF. Add 0 Ω resistors in series between the module and the USIM card connector to facilitate debugging. The 33 pF capacitors on USIM_DATA, USIM_CLK and USIM_RST are used for filtering out RF interference. In addition, keep the USIM peripheral circuit close to the USIM card connector.
- The pull-up resistor on USIM_DATA trace can improve anti-jamming capability when long layout trace and sensitive occasions are applied, and should be placed close to the USIM card connector.

4.5. PCM* and I2C* Interfaces

The module has one Pulse Code Modulation (PCM) digital interface and one I2C interface.

Table 17: Pin Definition of PCM and I2C Interfaces

Pin Name	Pin No.	I/O	Description	Comment
PCM_SYNC	31	DO	PCM data frame sync	If unused, keep them open.
PCM_CLK	30	DO	PCM clock	
PCM_DIN	32	DI	PCM data input	
PCM_DOUT	33	DO	PCM data output	
I2C_SCL	67	OD	I2C serial clock	External pull-up resistor is required. If unused, keep them open.
I2C_SDA	66	OD	I2C serial data	

A reference design of PCM and I2C interfaces with external Codec IC is illustrated in the following figure.

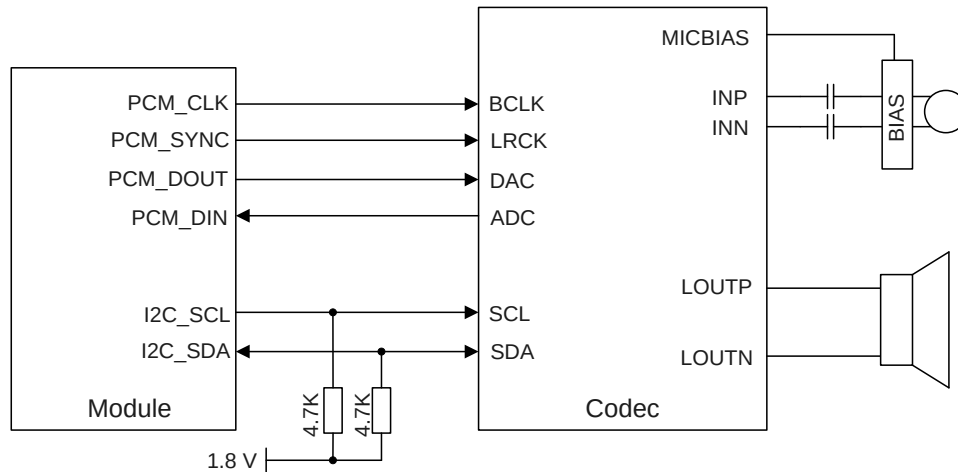


Figure 24: Reference Circuit of PCM and I2C Interfaces

NOTE

1. It is recommended to reserve RC circuits ($R = 22\ \Omega$, $C = 22\ \text{pF}$) on the PCM signal traces, especially on the PCM_CLK pin.
2. The module can only be used as a master device in applications related to both the PCM interface and the I2C interface.

4.6. ADC Interfaces

The module features two Analog-to-Digital Converter (ADC) interfaces. To improve ADC accuracy, the trace of ADC interfaces should be surrounded by ground.

Table 18: Pin Definition of ADC Interfaces

Pin Name	Pin No.	I/O	Description	Comment
ADC0	9	AI	General-purpose ADC interface	If unused, keep them open.
ADC1	96	AI	General-purpose ADC interface	

With **AT+QADC=<port>**, you can:

- **AT+QADC=0**: read the voltage value on ADC0
- **AT+QADC=1**: read the voltage value on ADC1

For more details about the AT command, see **document [2]**.

Table 19: ADC Interface Features

Name	Min.	Typ.	Max.	Unit
ADC input voltage range	0	-	1.2	V
ADC input resistance	0.26	-	0.75	MΩ
ADC resolution	-	12	-	bits

NOTE

If the acquisition voltage is greater than or equal to 1.2 V, it is recommended to use resistor divider circuit for ADC interface application. Resistance of the resistor divider should not exceed 100 kΩ, or the measurement accuracy of ADC would be significantly reduced. It is recommended to reserve a 100 nF capacitor for the design.

4.7. PSM_IND* & PSM_INT*

The module supports power saving mode (PSM).

Table 20: Pin Definition of PSM_IND and PSM_INT

Pin Name	Pin No.	I/O	Description	Comment
PSM_IND	83	DO	Indicate the module's power saving mode	If unused, keep it open.
PSM_INT	87	DI	External interrupt; wake up the module from power saving mode	Externally pulling up this pin can make the module exit power saving mode. If unused, keep it open.

4.8. Indication Signals

Table 21: Pin Definition of Indication Signals

Pin Name	Pin No.	I/O	Description	Comment
NET_STATUS	16	DO	Indicate the module's network activity status	If unused, keep them open.
STATUS	25	DO	Indicate the module's operation status	

4.8.1. Network Status Indication

The NET_STATUS pin indicates the module's network activity status, and can be used to drive network status indication LEDs.

Table 22: Network Connection Status/Activity Indicated by NET_STATUS Pin

Pin Name	Status	Description
NET_STATUS	Blink slowly (200 ms High/1800 ms Low)	Network searching
	Blink slowly (1800 ms High/200 ms Low)	Idle
	Blink quickly (125 ms High/125 ms Low)	Ongoing data transmission

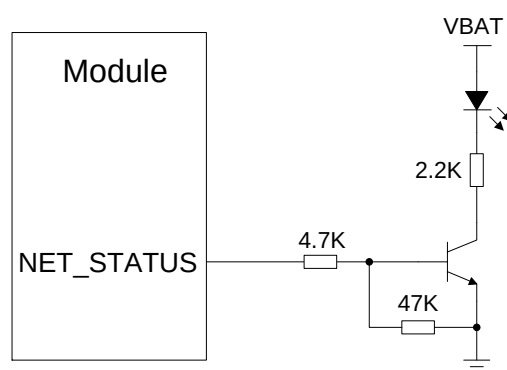


Figure 25: Reference Design of NET_STATUS

4.8.2. STATUS

The STATUS pin is used to indicate the module's operation status. It outputs high level when the module is turned on successfully.

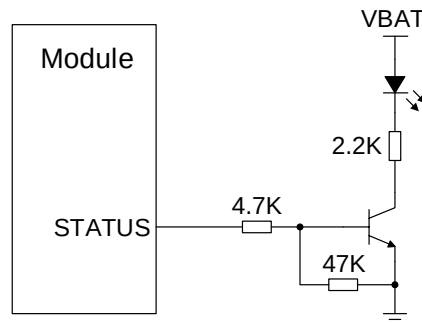


Figure 26: Reference Design of STATUS

4.8.3. MAIN_RI

AT+QCFG= "risignalttype","physical" can be used to configure MAIN_RI as URC indication pin. An URC will trigger MAIN_RI regardless of the URC output port. For more details about **AT+QCFG**, see *document [2]*.

NOTE

The **AT+QURCCFG** allows you to configure the main UART, USB AT port or USB modem port as the URC output port. The USB AT port is the URC output port by default. For more details about **AT+QURCCFG**, see *document [2]*.

MAIN_RI can be configured flexibly. However, its default behaviors are:

Table 23: Behaviors of MAIN_RI

Module Status	MAIN_RI Level Status
Idle	High
When a new URC information returns	MAIN_RI outputs at least 120 ms low level. After this pin changes to a high level, the module starts to output data.

The MAIN_RI can be configured as URC indication pin via **AT+QCFG="urc/ri/ring"**.

5 RF Specifications

Appropriate antenna type and design should be used with matched antenna parameters according to specific application. It is required to perform a comprehensive functional test for the RF design before mass production of terminal products. The entire content of this chapter is provided for illustration only. Analysis, evaluation and determination are still necessary when designing target products.

5.1. LTE/Wi-Fi Scan Antenna Interface

5.1.1. Antenna Interface & Frequency Bands

Table 24: Pin Description of LTE/Wi-Fi Scan Antenna Interface

Pin Name	Pin No.	I/O	Description	Comment
ANT_MAIN	35	AIO	Main antenna/Wi-Fi Scan antenna interface	50 Ω impedance.

NOTE

1. The Wi-Fi scan function utilizes the same antenna interface as the main antenna. Due to this shared interface, Time Division Multiplexing (TDM) is employed since the two functions cannot be used simultaneously. Wi-Fi Scan functionality only supports receiving. Transmitting is not supported.
2. Main antenna and Wi-Fi Scan antenna only support **passive antennas**.

Table 25: Operating Frequency

Operating Frequency	Transmit (MHz)	Receive (MHz)
LTE-FDD B1	1920–1980	2110–2170
LTE-FDD B2	1850–1910	1930–1990
LTE-FDD B3	1710–1785	1805–1880

LTE-FDD B4	1710–1755	2110–2155
LTE-FDD B5	824–849	869–894
LTE-FDD B7	2500–2570	2620–2690
LTE-FDD B8	880–915	925–960
LTE-FDD B12	699–716	729–746
LTE-FDD B13	777–787	746–756
LTE-FDD B14	788–798	758–768
LTE-FDD B17	704–716	734–746
LTE-FDD B18	815–830	860–875
LTE-FDD B19	830–845	875–890
LTE-FDD B20	832–862	791–821
LTE-FDD B25	1850–1915	1930–1995
LTE-FDD B26	814–849	859–894
LTE-FDD B28	703–748	758–803
LTE-TDD B34	2010–2025	2010–2025
LTE-TDD B38	2570–2620	2570–2620
LTE-TDD B39	1880–1920	1880–1920
LTE-TDD B40	2300–2400	2300–2400
LTE-TDD B41	2496–2690	2496–2690
LTE-FDD B66	1710–1780	2110–2180
LTE-FDD B71	663–698	617–652

5.1.2. Transmit Power

Table 26: Transmit Power

Frequency Bands	Max. RF Output Power	Min. RF Output Power
LTE	23 dBm $\pm$ 2 dB	< -39 dBm

NOTE

The module supports the above transmit power in hardware, but actual applications may be subject to local regulations.

5.1.3. Rx Sensitivity

Table 27: EG800Q-EU Conducted RF Receiver Sensitivity

Frequency Bands	Receiver Sensitivity (Typ.)	3GPP
LTE-FDD B1 (10 MHz)	-99.5 dBm	-93.3 dBm
LTE-FDD B3 (10 MHz)	-99.0 dBm	-90.3 dBm
LTE-FDD B5 (10 MHz)	-99.5 dBm	-91.8 dBm
LTE-FDD B7 (10 MHz)	-97.1 dBm	-91.3 dBm
LTE-FDD B8 (10 MHz)	-99.5 dBm	-90.8 dBm
LTE-FDD B20 (10 MHz)	-99.5 dBm	-90.8 dBm
LTE-FDD B28 (10 MHz)	-98.8 dBm	-92.3 dBm

Table 28: EG800Q-NA Conducted RF Receiver Sensitivity

Frequency Bands	Receiver Sensitivity (Typ.)	3GPP
LTE-FDD B2 (10 MHz)	-98.9 dBm	-91.3 dBm
LTE-FDD B4 (10 MHz)	-98.8 dBm	-93.3 dBm
LTE-FDD B5 (10 MHz)	-99.5 dBm	-91.8 dBm
LTE-FDD B12 (10 MHz)	-99.8 dBm	-90.3 dBm

LTE-FDD B13 (10 MHz)	-98.8 dBm	-90.3 dBm
LTE-FDD B66 (10 MHz)	-99.0 dBm	-92.8 dBm

Table 29: EG800Q-IN Conducted RF Receiver Sensitivity

Frequency Bands	Receiver Sensitivity (Typ.)	3GPP
LTE-FDD B1 (10 MHz)	-97.4 dBm	-93.3 dBm
LTE-FDD B3 (10 MHz)	-98 dBm	-90.3 dBm
LTE-FDD B5 (10 MHz)	-98.5 dBm	-91.8 dBm
LTE-FDD B8 (10 MHz)	-98 dBm	-90.8 dBm
LTE-TDD B40 (10 MHz)	-99.5 dBm	-93.8 dBm
LTE-TDD B41 (10 MHz)	-98 dBm	-91.8 dBm

Table 30: EG800Q-NAD Conducted RF Receiver Sensitivity

Frequency Bands	Receiver Sensitivity (Typ.)	3GPP
LTE-FDD B2 (10 MHz)	-98.9 dBm	-91.3 dBm
LTE-FDD B4 (10 MHz)	-98.8 dBm	-93.3 dBm
LTE-FDD B5 (10 MHz)	-99.5 dBm	-91.8 dBm
LTE-FDD B12 (10 MHz)	-99.8 dBm	-90.3 dBm
LTE-FDD B13 (10 MHz)	-98 dBm	-90.3 dBm
LTE-FDD B25 (10 MHz)	-98 dBm	-93.3 dBm
LTE-FDD B66 (10 MHz)	-99 dBm	-92.8 dBm
LTE-FDD B71 (10 MHz)	-97 dBm	-93.3 dBm

Table 31: EG800Q-GL Conducted RF Receiver Sensitivity

Frequency Bands	Receiver Sensitivity (Typ.)	3GPP
LTE-FDD B1 (10 MHz)	-98.85 dBm	-93.3 dBm

LTE-FDD B2 (10 MHz)	-99.49 dBm	-91.3 dBm
LTE-FDD B3 (10 MHz)	-98.75 dBm	-90.3 dBm
LTE-FDD B4 (10 MHz)	-98.95 dBm	-93.3 dBm
LTE-FDD B5 (10 MHz)	-98.12 dBm	-91.8 dBm
LTE-FDD B7 (10 MHz)	-98.19 dBm	-91.3 dBm
LTE-FDD B8 (10 MHz)	-96.62 dBm	-90.8 dBm
LTE-FDD B12 (10 MHz)	-98.19 dBm	-90.3 dBm
LTE-FDD B13 (10 MHz)	-97.75 dBm	-90.3 dBm
LTE-FDD B14 (10 MHz)	-97.82 dBm	-93.3 dBm
LTE-FDD B17 (10 MHz)	-98.22 dBm	-93.3 dBm
LTE-FDD B18 (10 MHz)	-97.92 dBm	-93.8 dBm
LTE-FDD B19 (10 MHz)	-97.89 dBm	-93.3 dBm
LTE-FDD B20 (10 MHz)	-98.62 dBm	-90.8 dBm
LTE-FDD B25 (10 MHz)	-99.45 dBm	-93.3 dBm
LTE-FDD B26 (10 MHz)	-98.15 dBm	-91.3 dBm
LTE-FDD B28 (10 MHz)	-98.35 dBm	-92.3 dBm
LTE-TDD B34 (10 MHz)	-99.42 dBm	-93.8 dBm
LTE-TDD B38 (10 MHz)	-98.85 dBm	-93.3 dBm
LTE-TDD B39 (10 MHz)	-99.82 dBm	-93.8 dBm
LTE-TDD B40 (10 MHz)	-99.22 dBm	-93.8 dBm
LTE-TDD B41 (10 MHz)	-98.72 dBm	-91.8 dBm
LTE-FDD B66 (10 MHz)	-98.62 dBm	-92.8 dBm
LTE-FDD B71 (10 MHz)	-99.12 dBm	-93.3 dBm

5.1.4. Reference Design

The module has one RF antenna interface for antenna connection. It is recommended to reserve a dual L-type circuit for better RF performance, and the dual L-type components (C3, C1, R1, and C2) should be placed as close as possible to the antenna. Capacitors C1 and C2 are not mounted by default.

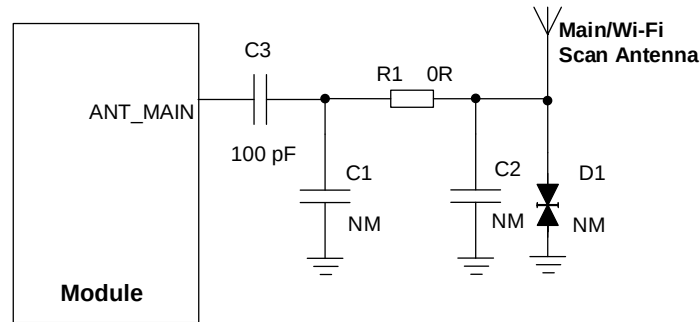


Figure 27: Reference Design of RF Antenna Interface

NOTE

1. Notes on C3:
 - 1) If there is DC power at the antenna ports, place a capacitor on C3 to prevent short circuit to ground. The capacitance value is recommended to be 100 pF, which can be adjusted according to the debugging results.
 - 2) If there is no DC power in the peripheral design:
 - a) Do not reserve C3.
 - b) If C3 has already been reserved, it should be mounted with a component, and it is recommended to use a 0 Ω resistor. You can also match the component according to the debugging results.
2. It is recommended to reserve an ESD protection component for the cellular antenna interface to effectively prevent static electricity and the junction capacitance should not exceed 0.05 pF.

5.2. RF Routing Guidelines

For user's PCB, the characteristic impedance of all RF traces should be controlled to 50 Ω . The impedance of the RF traces is usually determined by the trace width (W), the materials' dielectric constant, the height from the reference ground to the signal layer (H), and the spacing between RF traces and grounds (S). Microstrip or coplanar waveguide is typically used in RF layout to control characteristic impedance. The following are reference designs of microstrip or coplanar waveguide with different PCB structures.

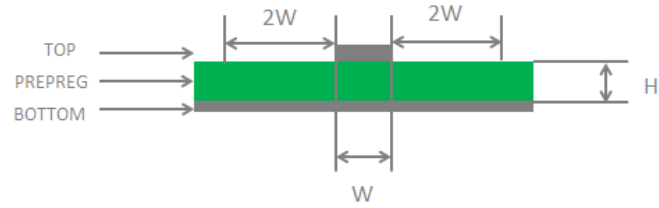


Figure 28: Microstrip Design on a 2-layer PCB

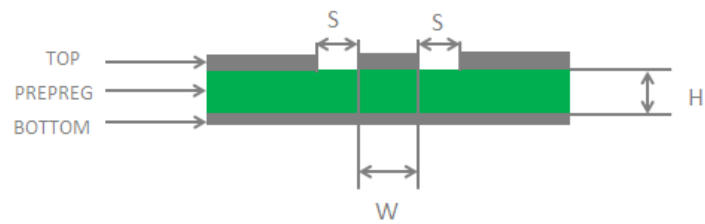


Figure 29: Coplanar Waveguide Design on a 2-layer PCB

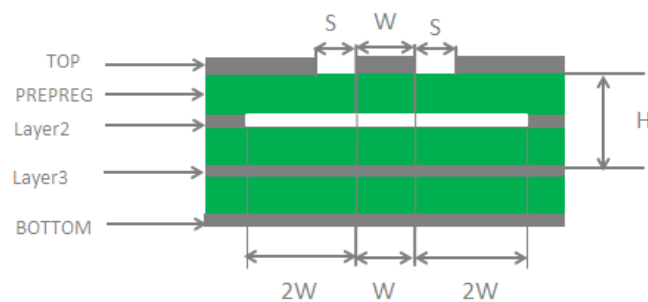


Figure 30: Coplanar Waveguide Design on a 4-layer PCB (Layer 3 as Reference Ground)

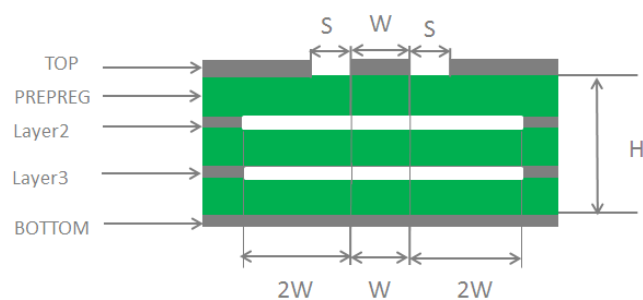


Figure 31: Coplanar Waveguide Design on a 4-layer PCB (Layer 4 as Reference Ground)

To ensure RF performance and reliability, follow the principles below in RF layout design:

- Use an impedance simulation tool to accurately control the characteristic impedance of RF traces to 50 Ω .
- The GND pins adjacent to RF pins should not be designed as thermal relief pads, and should be fully connected to ground.
- The distance between the RF pins and the RF connector should be as short as possible and all the right-angle traces should be changed to curved ones. The recommended trace angle is 135°.
- There should be clearance under the signal pin of the antenna connector or solder joint.
- The reference ground of RF traces should be complete. Meanwhile, adding some ground vias around RF traces and the reference ground could help to improve RF performance. The distance between the ground vias and RF traces should be at least twice the width of RF signal traces ($2 \times W$).
- Keep RF traces away from interference sources (such as DC-DC, (U)SIM/USB/SDIO high frequency digital signals, display signals, and clock signals), and avoid intersection and paralleling between traces on adjacent layers.

For more details about RF layout, see **document [4]**.

5.3. Antenna Design Requirements

Table 32: Antenna Design Requirements

Antenna Type	Requirements
Cellular/Wi-Fi Scan	● VSWR: ≤ 2 ● Efficiency: $> 30\%$ ● Gain: 1 dBi ● Max input power: 50 W ● Input impedance: 50 Ω ● Vertical polarization ● Cable insertion loss: < 1 dB: LB (< 1 GHz) < 1.5 dB: MB (1–2.3 GHz) < 2 dB: HB (> 2.3 GHz)

5.4. RF Connector Recommendation

If RF connector is used for antenna connection, it is recommended to use the U.FL-R-SMT connector provided by Hirose.

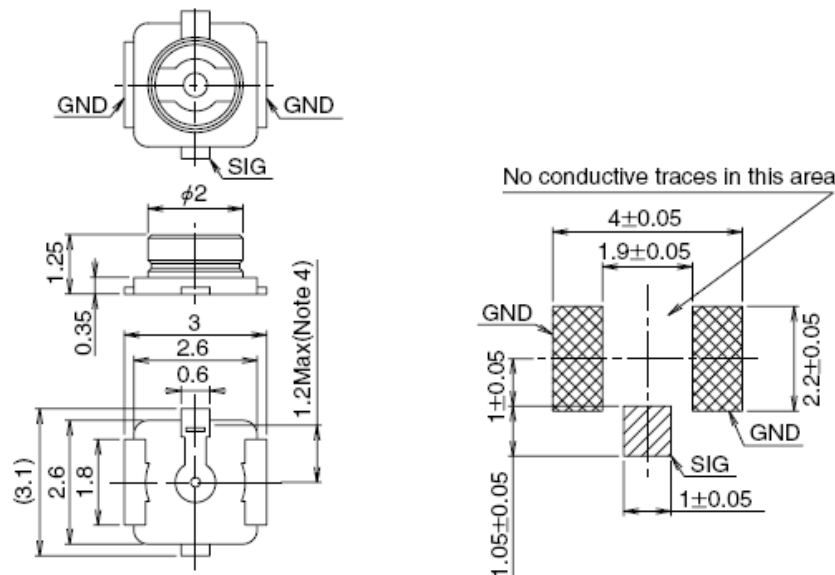


Figure 32: Receptacle Dimensions (Unit: mm)

U.FL-LP series mated plugs listed in the following figure can be used to match the U.FL-R-SMT connector.

Part No.	U.FL-LP-040	U.FL-LP-066	U.FL-LP(V)-040	U.FL-LP-062	U.FL-LP-088
Mated Height	2.5mm Max. (2.4mm Nom.)	2.5mm Max. (2.4mm Nom.)	2.0mm Max. (1.9mm Nom.)	2.4mm Max. (2.3mm Nom.)	2.4mm Max. (2.3mm Nom.)
Applicable cable	Dia. 0.81mm Coaxial cable	Dia. 1.13mm and Dia. 1.32mm Coaxial cable	Dia. 0.81mm Coaxial cable	Dia. 1mm Coaxial cable	Dia. 1.37mm Coaxial cable
Weight (mg)	53.7	59.1	34.8	45.5	71.7
RoHS	YES				

Figure 33: Specifications of Mated Plugs

The following figure describes the space factor of mated connectors.

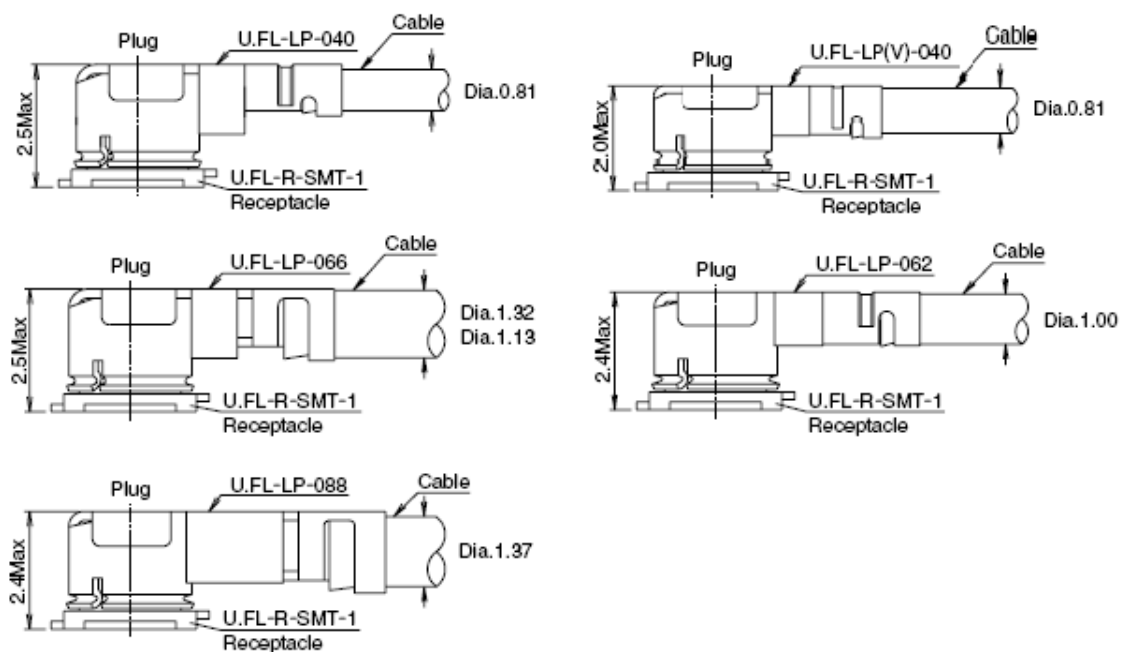


Figure 34: Space Factor of Mated Connectors (Unit: mm)

For more details, visit <http://www.hirose.com>.

6 Electrical Characteristics & Reliability

6.1. Absolute Maximum Ratings

Table 33: Absolute Maximum Ratings

Parameter	Min.	Max.	Unit
Voltage at VBAT	-0.3	5	V
Voltage at USB_VBUS	-0.3	5.25	V
Voltage at digital pins	-0.3	2.3	V

6.2. Power Supply Ratings

Table 34: Module Power Supply Ratings

Parameter	Description	Conditions	Min.	Typ.	Max.	Unit
VBAT	Power supply for the module	The actual input voltages must be kept between the minimum and maximum values.	3.3	3.8	4.3	V
I _{VBAT}	Peak supply current under BB and RF application scenarios	At maximum power control level	-	-	1.0	A
USB_VBUS	USB connection detect	-	3.0	5.0	5.25	V

6.3. Power Consumption

Table 35: EG800Q-EU Power Consumption

Test Item	Condition	Typ.	Unit
Turn-off	Turn off	55	μA
Sleep	AT+CFUN=0 (USB disconnected)	0.06	mA
	AT+CFUN=4 (USB disconnected)	0.15	mA
	LTE-FDD @ PF = 32 (USB disconnected)	1.23	mA
	LTE-FDD @ PF = 64 (USB disconnected)	0.70	mA
	LTE-FDD @ PF = 64 (USB suspended)	1.10	mA
	LTE-FDD @ PF = 128 (USB disconnected)	0.45	mA
	LTE-FDD @ PF = 256 (USB disconnected)	0.35	mA
	LTE-FDD @ eDRX Cycle Length 40.96 s PTW = 10.24 s PF = 32 (USB disconnected)	0.37	mA
	LTE-FDD @ eDRX Cycle Length 40.96 s PTW = 10.24 s PF = 64 (USB disconnected)	0.23	mA
	LTE-FDD @ eDRX Cycle Length 40.96 s PTW = 10.24 s PF = 128 (USB disconnected)	0.16	mA
	LTE-FDD @ eDRX Cycle Length 40.96 s PTW = 10.24 s PF = 256 (USB disconnected)	0.12	mA
Idle	LTE-FDD @ PF = 64 (USB disconnected)	4.50	mA
	LTE-FDD @ PF = 64 (USB connected)	25.50	mA
LTE data transmission	LTE-FDD B1	662	mA
	LTE-FDD B3	641	mA
	LTE-FDD B5	545	mA
	LTE-FDD B7	696	mA
	LTE-FDD B8	550	mA
	LTE-FDD B20	577	mA
	LTE-FDD B28	578	mA

Table 36: EG800Q-NA Power Consumption

Test Item	Condition	Typ.	Unit
Turn-off	Turn off	50	μA
Sleep	AT+CFUN=0 (USB disconnected)	0.06	mA
	AT+CFUN=4 (USB disconnected)	0.15	mA
	LTE-FDD @ PF = 32 (USB disconnected)	1.20	mA
	LTE-FDD @ PF = 64 (USB disconnected)	0.70	mA
	LTE-FDD @ PF = 64 (USB suspended)	1.00	mA
	LTE-FDD @ PF = 128 (USB disconnected)	0.40	mA
	LTE-FDD @ PF = 256 (USB disconnected)	0.30	mA
	LTE-FDD @ eDRX Cycle Length 40.96 s PTW = 10.24 s PF = 32 (USB disconnected)	0.38	mA
	LTE-FDD @ eDRX Cycle Length 40.96 s PTW = 10.24 s PF = 64 (USB disconnected)	0.24	mA
	LTE-FDD @ eDRX Cycle Length 40.96 s PTW = 10.24 s PF = 128 (USB disconnected)	0.17	mA
	LTE-FDD @ eDRX Cycle Length 40.96 s PTW = 10.24 s PF = 256 (USB disconnected)	0.13	mA
Idle	LTE-FDD @ PF = 64 (USB disconnected)	4.50	mA
	LTE-FDD @ PF = 64 (USB connected)	25.00	mA
LTE data transmission	LTE-FDD B2	588	mA
	LTE-FDD B4	656	mA
	LTE-FDD B5	499	mA
	LTE-FDD B12	572	mA
	LTE-FDD B13	586	mA
	LTE-FDD B66	631	mA

Table 37: EG800Q-IN Power Consumption

Test Item	Condition	Typ.	Unit
Turn-off	Turn off	1.54	μA
Sleep	AT+CFUN=0 (USB disconnected)	0.045	mA
	AT+CFUN=4 (USB disconnected)	0.13	mA
	LTE-FDD @ PF = 32 (USB disconnected)	1.16	mA
	LTE-FDD @ PF = 64 (USB disconnected)	0.64	mA
	LTE-FDD @ PF = 64 (USB suspended)	1.01	mA
	LTE-FDD @ PF = 128 (USB disconnected)	0.39	mA
	LTE-FDD @ PF = 256 (USB disconnected)	0.28	mA
	LTE-TDD @ PF = 32 (USB disconnected)	1.18	mA
	LTE-TDD @ PF = 64 (USB disconnected)	0.64	mA
	LTE-TDD @ PF = 128 (USB disconnected)	0.40	mA
	LTE-TDD @ PF = 256 (USB disconnected)	0.31	mA
Idle	LTE-FDD @ PF = 64 (USB disconnected)	4.23	mA
	LTE-FDD @ PF = 64 (USB connected)	24.79	mA
	LTE-TDD @ PF = 64 (USB disconnected)	3.90	mA
	LTE-TDD @ PF = 64 (USB connected)	24.82	mA
LTE data transmission	LTE-FDD B1	700	mA
	LTE-FDD B3	647	mA
	LTE-FDD B5	571	mA
	LTE-FDD B8	554	mA
	LTE-TDD B40	252	mA
	LTE-TDD B41	275	mA

Table 38: EG800Q-NAD Power Consumption

Test Item	Condition	Typ.	Unit
Turn-off	Turn off	50	μA
Sleep	AT+CFUN=0 (USB disconnected)	0.06	mA
	AT+CFUN=4 (USB disconnected)	0.14	mA
	LTE-FDD @ PF = 32 (USB disconnected)	1.20	mA
	LTE-FDD @ PF = 64 (USB disconnected)	0.70	mA
	LTE-FDD @ PF = 64 (USB suspended)	1.00	mA
	LTE-FDD @ PF = 128 (USB disconnected)	0.40	mA
	LTE-FDD @ PF = 256 (USB disconnected)	0.30	mA
	LTE-FDD @ eDRX Cycle Length 40.96 s PTW = 10.24 s PF = 32 (USB disconnected)	0.38	mA
	LTE-FDD @ eDRX Cycle Length 40.96 s PTW = 10.24 s PF = 64 (USB disconnected)	0.24	mA
	LTE-FDD @ eDRX Cycle Length 40.96 s PTW = 10.24 s PF = 128 (USB disconnected)	0.17	mA
	LTE-FDD @ eDRX Cycle Length 40.96 s PTW = 10.24 s PF = 256 (USB disconnected)	0.13	mA
Idle	LTE-FDD @ PF = 64 (USB disconnected)	4.77	mA
	LTE-FDD @ PF = 64 (USB connected)	24.98	mA
LTE data transmission	LTE-FDD B2	588	mA
	LTE-FDD B4	656	mA
	LTE-FDD B5	499	mA
	LTE-FDD B12	572	mA
	LTE-FDD B13	603	mA
	LTE-FDD B25	614	mA
	LTE-FDD B66	631	mA
	LTE-FDD B71	527	mA

Table 39: EG800Q-GL Power Consumption

Test Item	Condition	Typ.	Unit
Turn-off	Turn off	0.95	μA
PSM	Power Saving Mode	68.59	μA
Sleep	AT+CFUN=0 (USB disconnected)	0.54	mA
	AT+CFUN=4 (USB disconnected)	0.13	mA
	LTE-FDD @ PF = 32 (USB disconnected)	1.36	mA
	LTE-FDD @ PF = 64 (USB disconnected)	0.63	mA
	LTE-FDD @ PF = 128 (USB disconnected)	0.365	mA
	LTE-FDD @ PF = 256 (USB disconnected)	0.24	mA
	LTE-TDD @ PF = 32 (USB disconnected)	1.21	mA
	LTE-TDD @ PF = 64 (USB disconnected)	0.62	mA
	LTE-TDD @ PF = 128 (USB disconnected)	0.44	mA
	LTE-TDD @ PF = 256 (USB disconnected)	0.32	mA
	LTE-FDD @ eDRX Cycle Length 40.96 s PTW = 10.24 s PF = 32 (USB disconnected)	0.32	mA
	LTE-FDD @ eDRX Cycle Length 40.96 s PTW = 10.24 s PF = 64 (USB disconnected)	0.16	mA
	LTE-FDD @ eDRX Cycle Length 40.96 s PTW = 10.24 s PF = 128 (USB disconnected)	0.07	mA
	LTE-FDD @ eDRX Cycle Length 40.96 s PTW = 10.24 s PF = 256 (USB disconnected)	0.05	mA
	LTE-TDD @ eDRX Cycle Length 40.96 s PTW = 10.24 s PF = 32 (USB disconnected)	0.36	mA
	LTE-TDD @ eDRX Cycle Length 40.96 s PTW = 10.24 s PF = 64 (USB disconnected)	0.17	mA
	LTE-TDD @ eDRX Cycle Length 40.96 s PTW = 10.24 s PF = 128 (USB disconnected)	0.12	mA
	LTE-TDD @ eDRX Cycle Length 40.96 s PTW = 10.24 s PF = 256 (USB disconnected)	0.05	mA
	LTE-FDD @ PF = 64 (USB disconnected)	3.88	mA
	LTE-FDD @ PF = 64 (USB connected)	24.89	mA
	LTE-TDD @ PF = 64 (USB disconnected)	3.97	mA

LTE data transmission	LTE-TDD @ PF = 64 (USB connected)	24.89	mA
	LTE-FDD B1	621	mA
	LTE-FDD B2	665	mA
	LTE-FDD B3	664	mA
	LTE-FDD B4	675	mA
	LTE-FDD B5	623	mA
	LTE-FDD B7	763	mA
	LTE-FDD B8	640	mA
	LTE-FDD B12	604	mA
	LTE-FDD B13	686	mA
	LTE-FDD B14	599	mA
	LTE-FDD B17	587	mA
	LTE-FDD B18	567	mA
	LTE-FDD B19	556	mA
	LTE-FDD B20	649	mA
	LTE-FDD B25	670	mA
	LTE-FDD B26	614	mA
	LTE-FDD B28	639	mA
	LTE-TDD B34	294	mA
	LTE-TDD B38	281	mA
	LTE-TDD B39	299	mA
	LTE-TDD B40	295	mA
	LTE-TDD B41	300	mA
	LTE-FDD B66	674	mA
	LTE-FDD B71	624	mA

NOTE

The above power consumption data are tested under 50 Ω impedance.

6.4. Digital I/O Characteristics

Table 40: VDD_EXT I/O Requirements

Parameter	Description	Min.	Max.	Unit
V _{IH}	High-level input voltage	$0.7 \times VDD_EXT$	$VDD_EXT + 0.3$	V
V _{IL}	Low-level input voltage	-0.3	$0.2 \times VDD_EXT$	V
V _{OH}	High-level output voltage	$0.8 \times VDD_EXT$	-	V
V _{OL}	Low-level output voltage	-	$0.15 \times VDD_EXT$	V

Table 41: USIM Low/High-voltage I/O Requirements

Parameter	Description	Min.	Max.	Unit
V _{IH}	High-level input voltage	$0.7 \times USIM_VDD$	-	V
V _{IL}	Low-level input voltage	-	$0.2 \times USIM_VDD$	V
V _{OH}	High-level output voltage	$0.8 \times USIM_VDD$	-	V
V _{OL}	Low-level output voltage	-	$0.15 \times USIM_VDD$	V

6.5. ESD Protection

Static electricity occurs naturally and may damage the module. Therefore, applying proper ESD countermeasures and handling methods is imperative. For example, wear anti-static gloves during the development, production, assembly and testing of the module; add ESD protection components to the ESD sensitive interfaces and points in the product design.

Table 42: Electrostatics Discharge Characteristics (Temperature: 15–35 °C, Humidity: 30–60 %)

Tested Interfaces	Contact Discharge	Air Discharge	Unit
VBAT, GND	±5	±10	kV
Antenna Interface	±4	±8	kV
Other Interfaces	±0.5	±1	kV

6.6. Operating and Storage Temperatures

Table 43: Operating and Storage Temperatures

Parameter	Min.	Typ.	Max.	Unit
Normal operating temperature ⁴	-35	+25	+75	°C
Extended operating temperature ⁵	-40	-	+85	°C
Storage temperature	-40	-	+90	°C

⁴ Within this range, the module's indicators comply with 3GPP specification requirements.

⁵ Within the range of -40 to -35 °C or 75 to 85 °C, the module retains the ability to establish and maintain functions such as SMS and data transmission, without any unrecoverable malfunction. Radio spectrum and radio network remain uninfluenced, whereas the value of one or more parameters, such as P_{out}, may decrease and fall below the range of the 3GPP specified tolerances. When the temperature returns to the normal operating temperature range, the module's indicators will comply with 3GPP specification requirements again.

7 Mechanical Information

This chapter describes the mechanical dimensions of the module. All dimensions are measured in millimeters (mm), and the dimensional tolerances are ± 0.2 mm unless otherwise specified.

7.1. Mechanical Dimensions

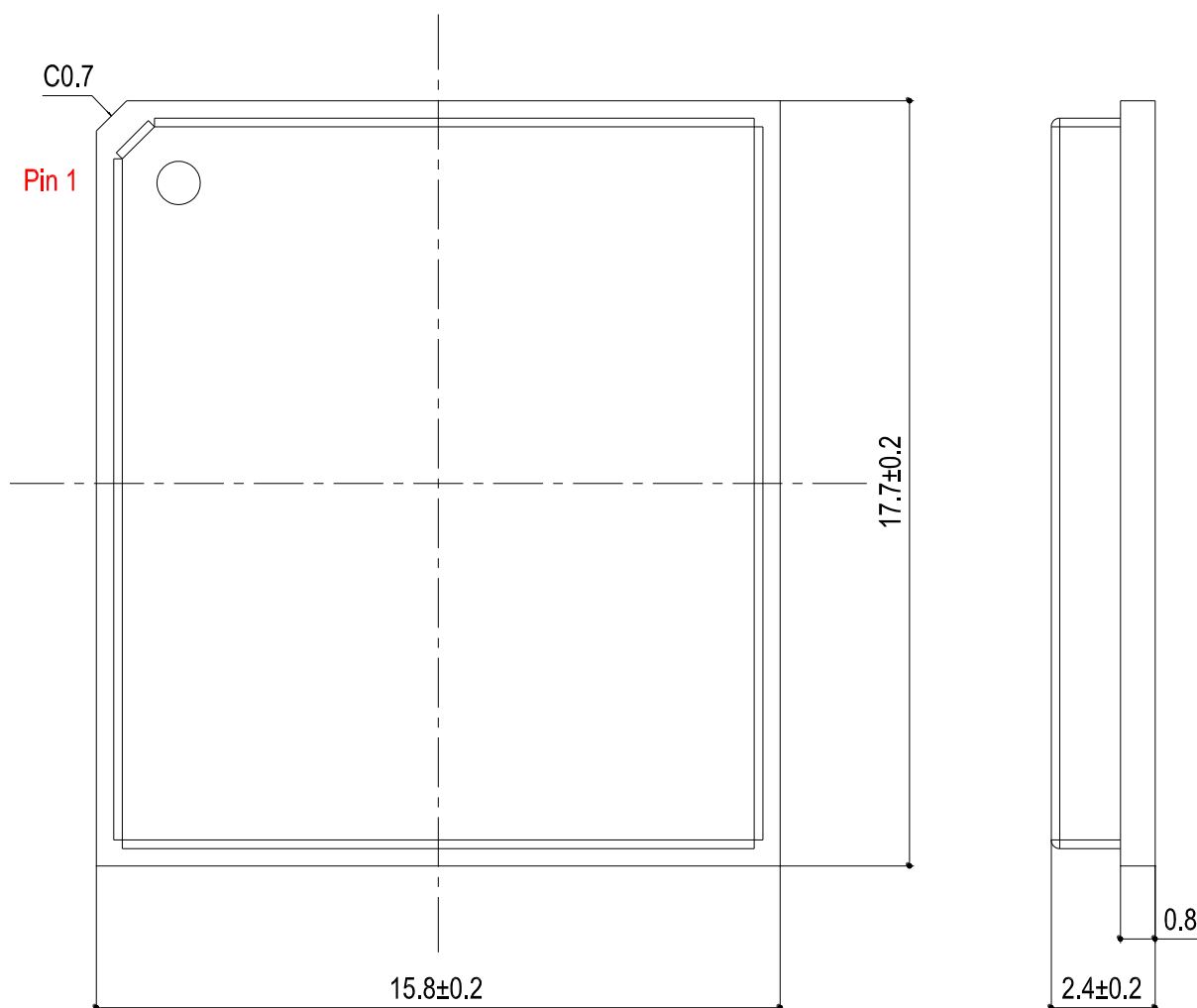


Figure 35: Module Top and Side Dimensions (Unit: mm)

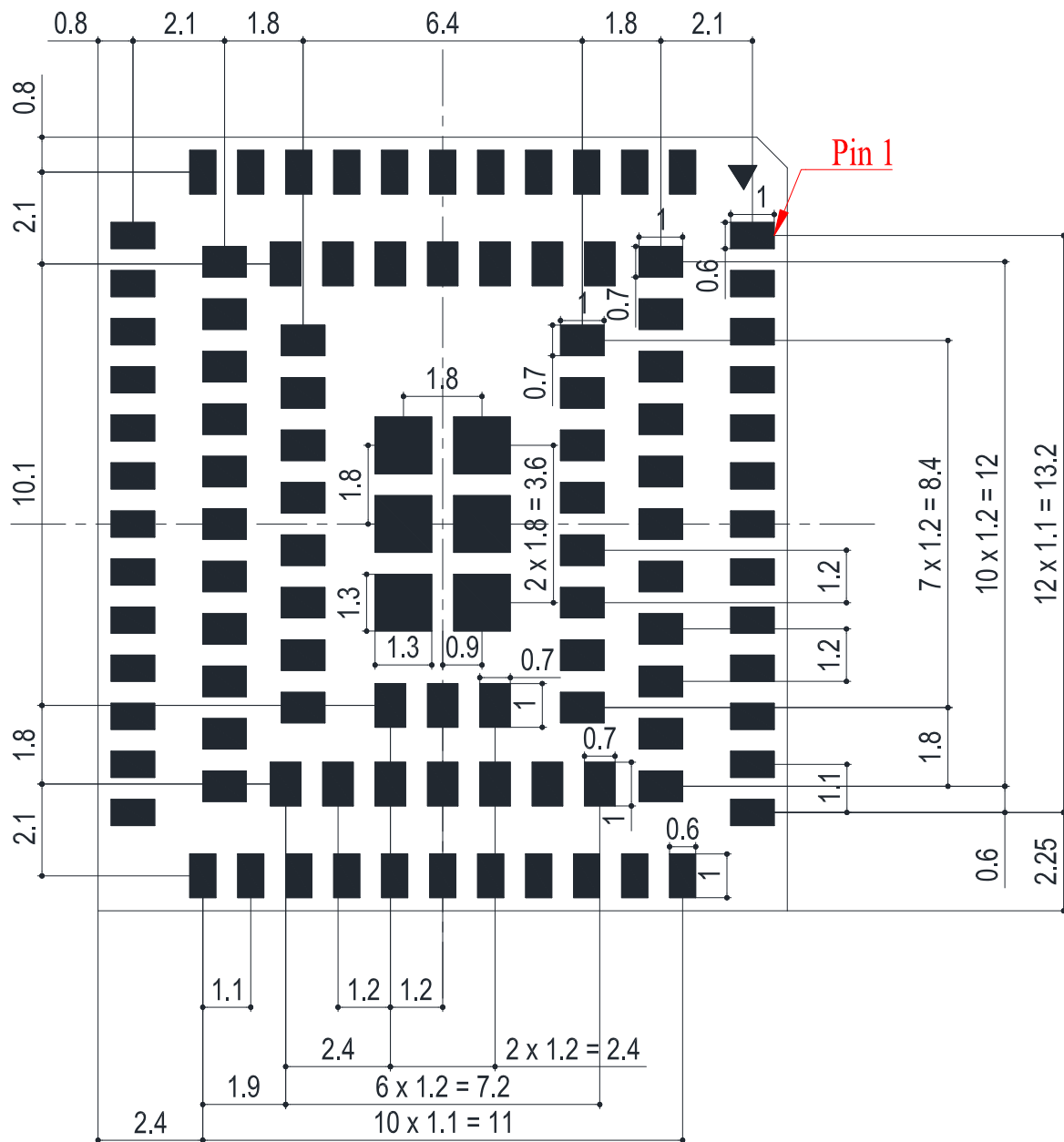


Figure 36: Bottom Dimensions (Bottom View, Unit: mm)

NOTE

The module's coplanarity standard: ≤ 0.13 mm.

7.2. Recommended Footprint

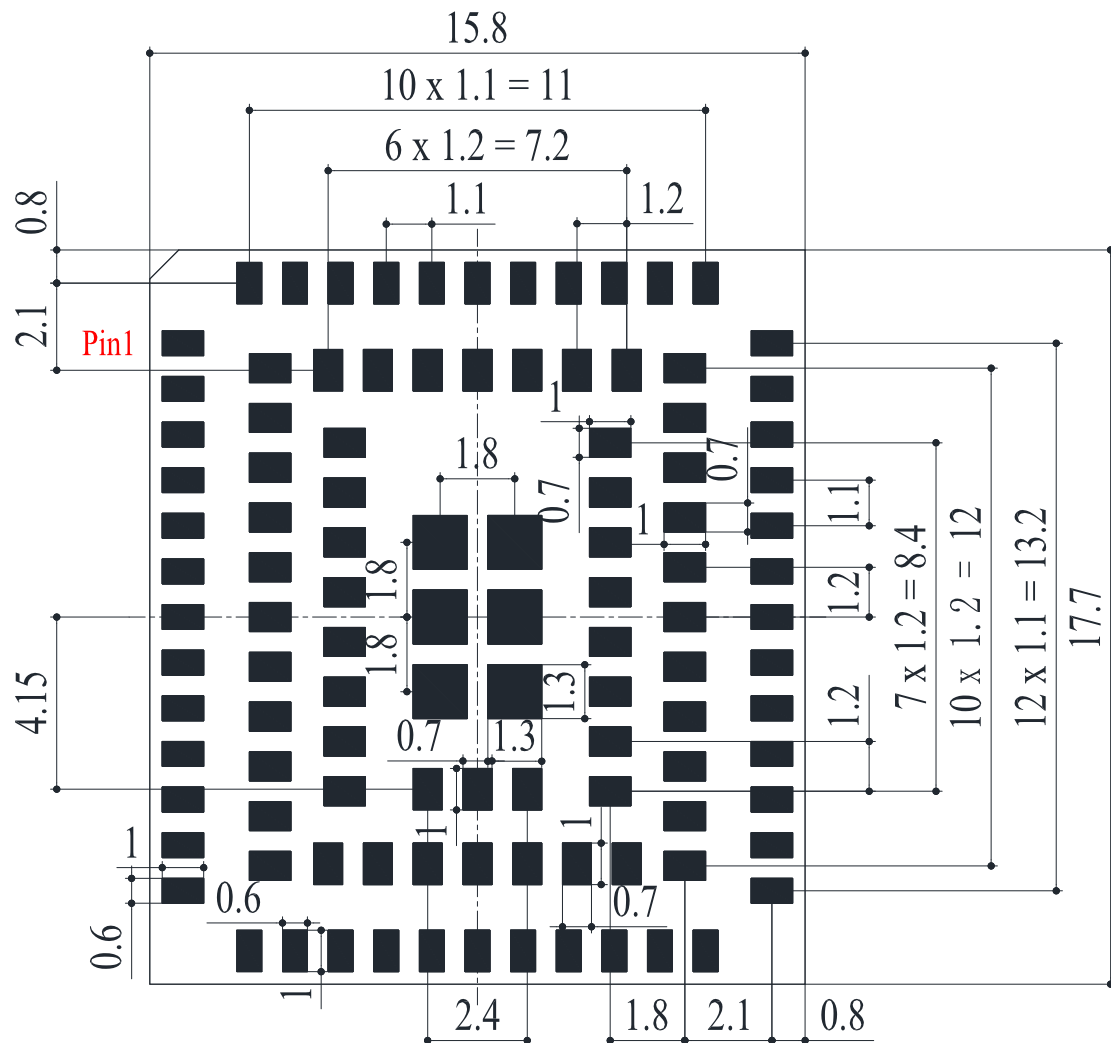


Figure 37: Recommended Footprint (Unit: mm)

NOTE

Keep at least 3 mm between the module and other components on the motherboard to improve soldering quality and maintenance convenience.

7.3. Top and Bottom Views

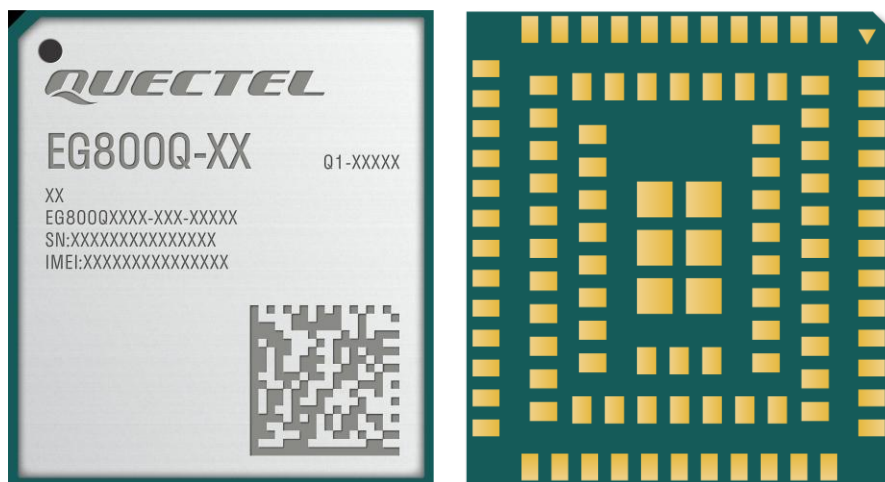


Figure 38: EG800Q-EU & EG800Q-NA & EG800Q-IN & EG800Q-GL Module Top and Bottom Views

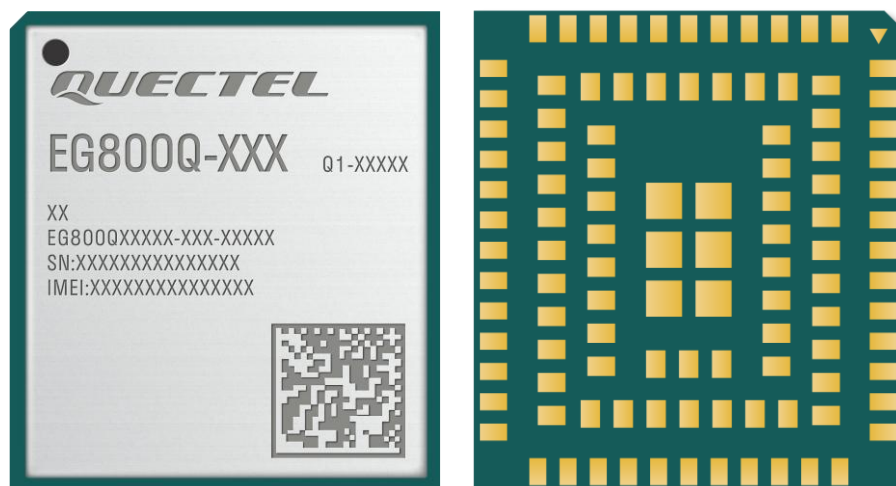


Figure 39: EG800Q-NAD Module Top and Bottom Views

NOTE

Images above are for illustrative purposes only and may differ from the actual module. For authentic appearance and label, please refer to the module received from Quectel.

8 Storage, Manufacturing & Packaging

8.1. Storage Conditions

The module is provided with vacuum-sealed packaging. MSL of the module is rated as 3. The storage requirements are shown below.

1. Recommended Storage Condition: the temperature should be 23 ± 5 °C and the relative humidity should be 35–60 %.
2. Shelf life (in a vacuum-sealed packaging): 12 months in Recommended Storage Condition.
3. Floor life: 168 hours ⁶ in a factory where the temperature is 23 ± 5 °C and relative humidity is below 60 %. After the vacuum-sealed packaging is removed, the module must be processed in reflow soldering or other high-temperature operations within 168 hours. Otherwise, the module should be stored in an environment where the relative humidity is less than 10 % (e.g., a dry cabinet).
4. The module should be pre-baked to avoid blistering, cracks and inner-layer separation in PCB under the following circumstances:
 - The module is not stored in Recommended Storage Condition;
 - Violation of the third requirement mentioned above;
 - Vacuum-sealed packaging is broken, or the packaging has been removed for over 24 hours;
 - Before module repairing.
5. If needed, the pre-baking should follow the requirements below:
 - The module should be baked for 24 hours at 120 ± 5 °C;
 - The module must be soldered to PCB within 24 hours after the baking, otherwise it should be put in a dry environment such as in a dry cabinet.

⁶ This floor life is only applicable when the environment conforms to *IPC/JEDEC J-STD-033*. It is recommended to start the solder reflow process within 24 hours after the package is removed if the temperature and moisture do not conform to, or are not sure to conform to *IPC/JEDEC J-STD-033*. And do not unpack the modules in large quantities until they are ready for soldering.

NOTE

1. To avoid blistering, layer separation and other soldering issues, extended exposure of the module to the air is forbidden.
2. Take out the module from the package and put it on high-temperature-resistant fixtures before baking. If shorter baking time is desired, see *IPC/JEDEC J-STD-033* for the baking procedure.
3. Pay attention to ESD protection, such as wearing anti-static gloves, when touching the modules.

8.2. Manufacturing and Soldering

Push the squeegee to apply the solder paste on the surface of stencil, thus making the paste fill the stencil openings and then penetrate to the PCB. Apply proper force on the squeegee to produce a clean stencil surface on a single pass. To guarantee module soldering quality, the thickness of stencil for the module is recommended to be 0.13–0.15 mm. For more details, see **document [5]**.

The recommended peak reflow temperature should be 235–246 °C, with 246 °C as the absolute maximum reflow temperature. To avoid damage to the module caused by repeated heating, it is recommended that the module should be mounted only after reflow soldering for the other side of PCB has been completed. The recommended reflow soldering thermal profile (lead-free reflow soldering) and related parameters are shown below.

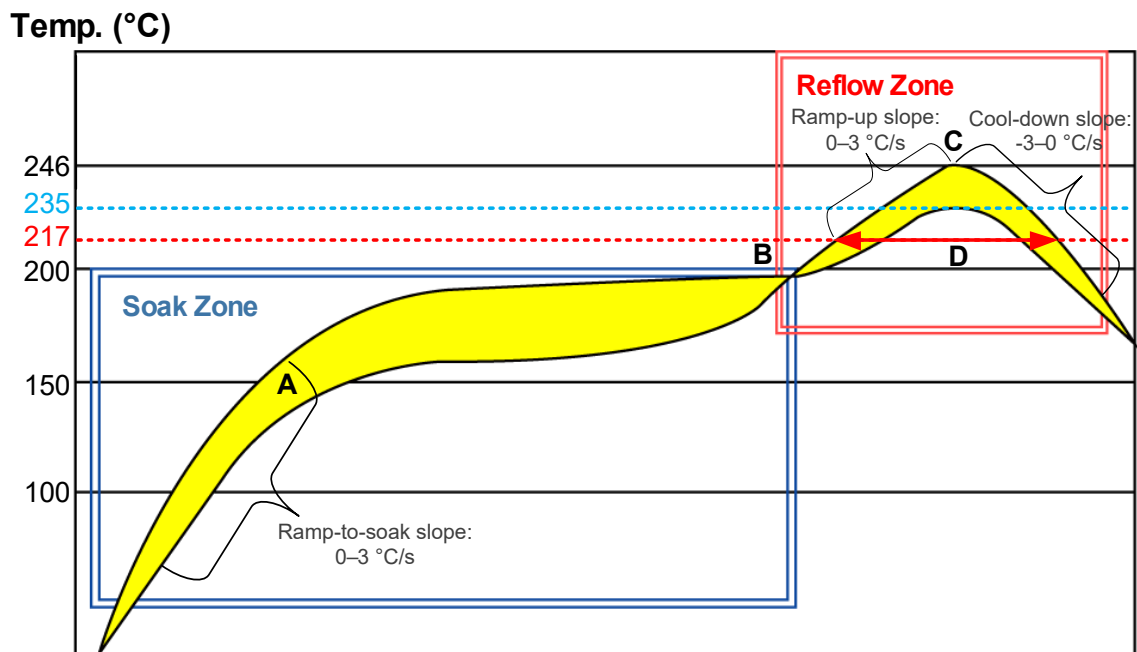


Figure 40: Recommended Reflow Soldering Thermal Profile

Table 44: Recommended Thermal Profile Parameters

Factor	Recommended Value
Soak Zone	
Ramp-to-soak Slope	0–3 °C/s
Soak Time (between A and B: 150 °C and 200 °C)	70–120 s
Reflow Zone	
Ramp-up Slope	0–3 °C/s
Reflow Time (D: over 217 °C)	40–70 s
Max Temperature	235–246 °C
Cool-down Slope	-3–0 °C/s
Reflow Cycle	
Max Reflow Cycle	1

NOTE

1. The above profile parameter requirements are for the measured temperature of the solder joints. Both the hottest and coldest spots of solder joints on the PCB should meet the above requirements.
2. During manufacturing and soldering, or any other processes that may contact the module directly, **NEVER** wipe the module's shielding can with organic solvents, such as acetone, ethyl alcohol, isopropyl alcohol, trichloroethylene. Otherwise, the shielding can may become rusted.
3. The shielding can for the module is made of Cupro-Nickel base material. It is tested that after 12 hours' Neutral Salt Spray test, the laser engraved label information on the shielding can is still clearly identifiable and the QR code is still readable, although white rust may be found.
4. If a conformal coating is necessary for the module, do **NOT** use any coating material that may chemically react with the PCB or shielding cover, and prevent the coating material from flowing into the module.
5. Avoid using ultrasonic technology for module cleaning since it can damage crystals inside the module.
6. Avoid using materials that contain mercury (Hg), such as adhesives, for module processing, even if the materials are RoHS compliant and their mercury content is below 1000 ppm (0.1 %).
7. Corrosive gases may corrode the electronic components inside the module, affecting their reliability and performance, and potentially leading to a shortened service life that fails to meet the designed lifespan. Therefore, do not store or use unprotected modules in environments containing corrosive gases such as hydrogen sulfide, sulfur dioxide, chlorine, and ammonia.
8. Due to the complexity of the SMT process, please contact Quectel Technical Support in advance for any situation that you are not sure about, or any process (e.g. selective soldering, ultrasonic

soldering) that is not mentioned in **document [6]**.

8.3. Packaging Specifications

This chapter outlines the key packaging parameters and processes. All figures below are for reference purposes only, as the actual appearance and structure of packaging materials may vary in delivery.

The modules are packed in a tape and reel packaging as specified in the sub-chapters below.

8.3.1. Carrier Tape

Carrier tape dimensions are illustrated in the following figure and table:

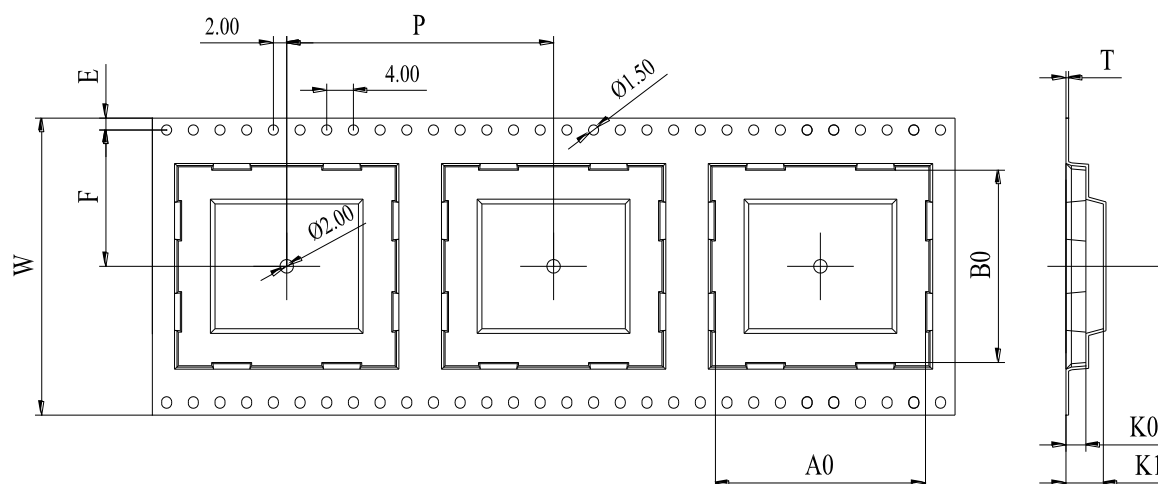


Figure 41: Carrier Tape Dimension Drawing (Unit: mm)

Table 45: Carrier Tape Dimension Table (Unit: mm)

W	P	T	A0	B0	K0	K1	F	E
32	24	0.4	16.2	18.1	2.8	4.6	14.2	1.75

8.3.2. Plastic Reel

Plastic reel dimensions are illustrated in the following figure and table:

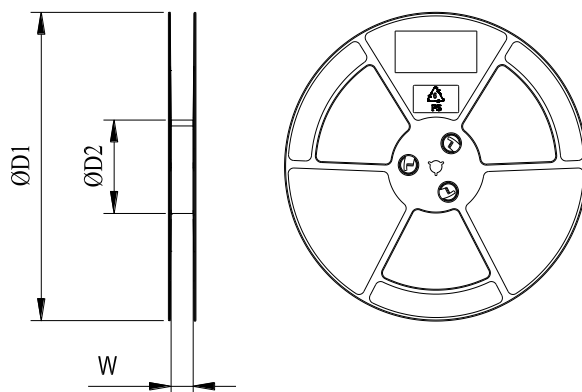


Figure 42: Plastic Reel Dimension Drawing

Table 46: Plastic Reel Dimension Table (Unit: mm)

ØD1	ØD2	W
330	100	32.5

8.3.3. Mounting Direction

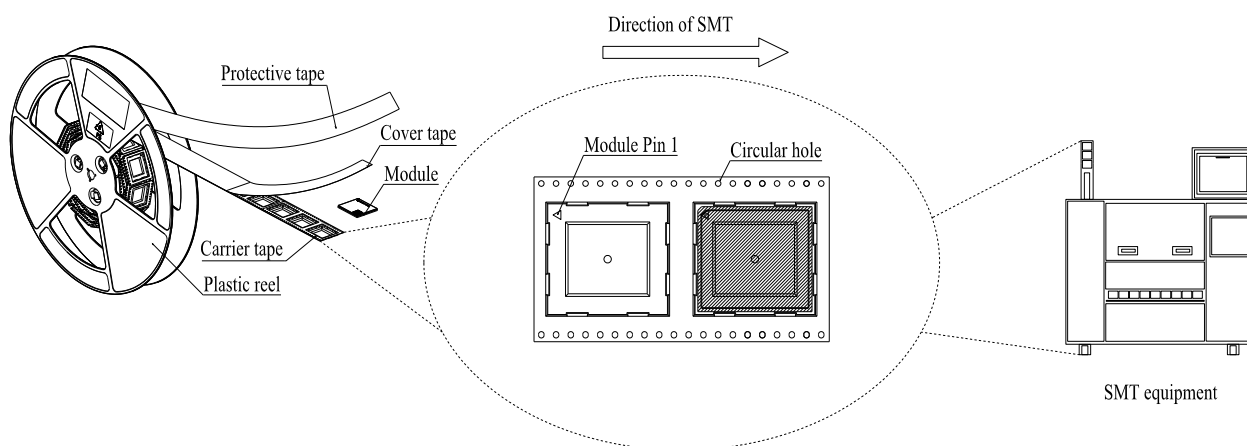
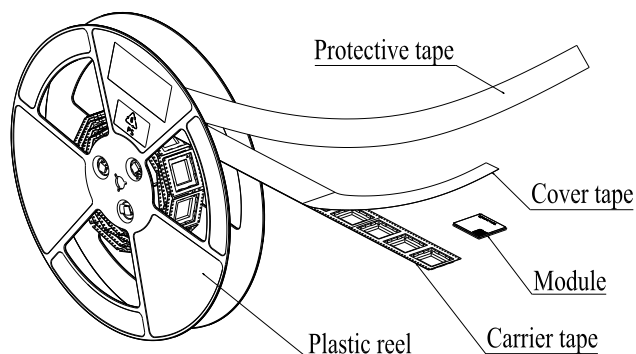


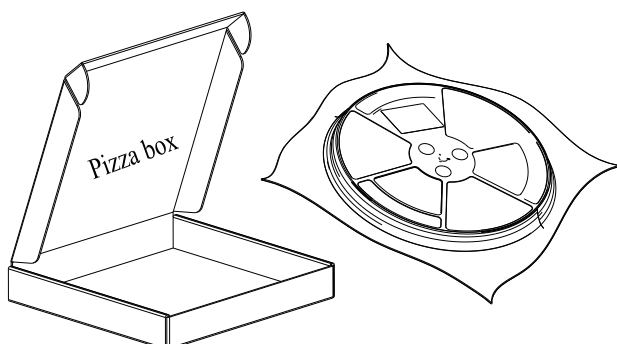
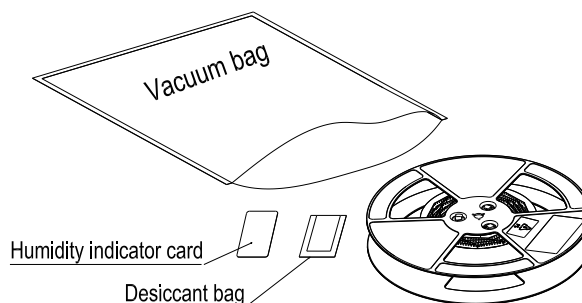
Figure 43: Mounting Direction

8.3.4. Packaging Process



Place the modules onto the carrier tape cavity and cover them securely with cover tape. Wind the heat-sealed carrier tape onto a plastic reel and apply a protective tape for additional protection. 1 plastic reel can pack 500 modules.

Place the packaged plastic reel, humidity indicator card and desiccant bag into a vacuum bag, and vacuumize it.



Place the vacuum-packed plastic reel into a pizza box.

Place the 4 packaged pizza boxes into 1 carton and seal it. 1 carton can pack 2000 modules.

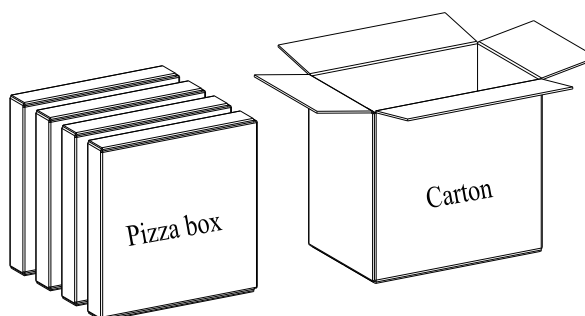


Figure 44: Packaging Process

9 Appendix References

Table 47: Related Documents

Document Name
[1] Quectel_UMTS<E_EVB_User_Guide
[2] Quectel_EG800Q&EG91xQ_Series_AT_Commands_Manual
[3] Quectel_EG800Q&EG91xQ_Series_PSM&eDRX_Application_Note
[4] Quectel_RF_Layout_Application_Note
[5] Quectel_Module_Stencil_Design_Requirements
[6] Quectel_Module_SMT_Application_Note

Table 48: Terms and Abbreviations

Abbreviation	Description
3GPP	3rd Generation Partnership Project
bps	Bytes per second
CHAP	Challenge Handshake Authentication Protocol
CMUX	Connection MUX
CTS	Clear To Send
DFOTA	Delta Firmware Upgrade Over-The-Air
DSSS	Direct Sequence Spread Spectrum
EMI	Electromagnetic Interference
ESD	Electrostatic Discharge
ESR	Equivalent Series Resistance

ETSI	European Telecommunications Standards Institute
EVB	Evaluation Board
FDD	Frequency Division Duplexing
FILE	File Protocol
FTP	File Transfer Protocol
FTPS	FTP over SSL
GND	Ground
HB	High Band
HTTP	Hypertext Transfer Protocol
HTTPS	Hypertext Transfer Protocol Secure
IMT-2000	International Mobile Telecommunications 2000
I2C	Inter-Integrated Circuit
I _o max	Maximum Output Load Current
LB	Low Band
LCC	Leadless Chip Carrier (package)
LDO	Low-dropout Regulator
LED	Light Emitting Diode
LGA	Land Grid Array
LTE	Long Term Evolution
MB	Medium Band
MCU	Microcontroller Unit/Microprogrammed Control Unit
ME	Mobile Equipment
MMS	Multimedia Messaging Service
MO	Mobile Origination
MQTT	Message Queuing Telemetry Transport

MT	Mobile Termination
NITZ	Network Identity and Time Zone
NTP	Network Time Protocol
PAP	Password Authentication Protocol
PCB	Printed Circuit Board
PCM	Pulse Code Modulation
PDU	Protocol Data Unit
PING	Packet Internet Groper
PPP	Point-to-Point Protocol
QAM	Quadrature Amplitude Modulation
QPSK	Quadrature Phase Shift Keying
RF	Radio Frequency
RoHS	Restriction of Hazardous Substances
RTS	Request To Send
SMD	Surface Mount Device
SMS	Short Message Service
SMTP	Simple Mail Transfer Protocol
SMTPS	Simple Mail Transfer Protocol Secure
SSL	Secure Sockets Layer
TCP	Transmission Control Protocol
TVS	Transient Voltage Suppressor
UART	Universal Asynchronous Receiver/Transmitter
UDP	User Datagram Protocol
UMTS	Universal Mobile Telecommunications System
URC	Unsolicited Result Code

USB	Universal Serial Bus
USIM	Universal Subscriber Identity Module
VBAT	Voltage at Battery (Pin)
V _{IH}	High-level input voltage
V _{IL}	Low-level input voltage
V _{OH}	High-level output voltage
V _{OL}	Low-level output voltage
V _{max}	Maximum Voltage
V _{min}	Minimum Voltage
V _{nom}	Nominal Voltage
V _{ILmax}	Maximum Low-level Input Voltage
V _{RWM}	Working Peak Reverse Voltage
VSWR	Voltage Standing Wave Ratio
